The impact of punctuated infectiousness on infectious disease dynamics

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Abstract

1 Introduction

Epidemic trajectories are profoundly shaped by individual-level effects. For example, interpersonal variation in pathogen shedding and contact rates can lead to superspreading, which impacts both the likelihood that an epidemic will occur and how explosive it will be if it does [Lloyd-Smith et al., 2005]. Likewise, chance events early in an epidemic can modulate when widespread transmission becomes established, durably shifting the outbreak's overall timing [Morris et al., 2024]. Accounting for such effects is vital for public health preparedness: while understanding an epidemic's expected trajectory is useful — and can be done by largely overlooking individual-level effects — assessing variation around that mean, including reasonable best- and worst-case scenarios, requires careful attention to individual-level transmission dynamics.

The mean-field dynamics of an epidemic are determined by two fundamental quantities: the basic reproduction number, R_0, and the intrinsic generation interval distribution, $g(\tau)$ [Wallinga and Lipsitch, 2007], where τ represents time since infection. These can be combined into a single object, the "infectiousness profile" $A(\tau) = R_0\, g(\tau)$, which captures both the expected amount of transmission arising from an index case and its timing, and is the key parameter in the renewal equation framework for epidemic modeling [Breda et al., 2012]. Analogously, at the individual level, the individual reproduction number, ν_i, captures the expected number of infections person i would produce in an otherwise susceptible population [Lloyd-Smith et al., 2005], and the individual-level intrinsic generation interval distribution, $g_i(\tau)$, captures the timing of those infections. These can be combined into the "individual infectiousness profile" or the "infection kernel", $a_i(\tau) = \nu_i g_i(\tau)$ [Champredon and Dushoff, 2015, Svensson, 2007]. The population-level quantities are the expectations of the individual-level ones, with $R_0 = E_i[\nu_i]$, $g(\tau) = E_i[g_i(\tau)]$, and $A(\tau) = E_i[a_i(\tau)]$. Substantial work has gone into understanding how variation in ν_i (*i.e.*, the area under $a_i(\tau)$) impacts epidemic dynamics, but the impact of variation in $g_i(\tau)$ (*i.e.*, how $a_i(\tau)$ is distributed over time), holding R_0 and $g(\tau)$ fixed and without confounding from variation in ν_i, is less well understood.

Since $A(\tau)$ is an expectation over individuals, it need not resemble any single person's $a_i(\tau)$. The standard susceptible-exposed-infectious-recovered (SEIR) model provides a useful illustration: it assumes each person undergoes a latent period of length $x_i \sim \mathrm{Exp}(\gamma_{\mathrm{SEIR}})$ during which they are not infectious, followed by an infectious period of length $y_i \sim \mathrm{Exp}(\alpha_{\mathrm{SEIR}})$ during which they transmit at rate β_{SEIR}. The resulting "stepwise" individual infectiousness profile is

$$a_i(\tau) = \begin{cases} \beta_{\mathrm{SEIR}} & \tau \in [x_i, x_i + y_i] \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Averaging over individuals yields the population-level infectiousness profile

$$A_{\mathrm{SEIR}}(\tau) = E_i[a_i(\tau)] = \beta_{\mathrm{SEIR}} \frac{\gamma_{\mathrm{SEIR}}}{\gamma_{\mathrm{SEIR}} - \alpha_{\mathrm{SEIR}}} (e^{-\alpha_{\mathrm{SEIR}}\tau} - e^{-\gamma_{\mathrm{SEIR}}\tau}) \tag{2}$$

i.e., a smooth, bell-shaped curve that emerges from a collection of discontinuous $a_i(\tau)$ (**Figure XX**). Other choices of $a_i(\tau)$ can also produce the same $A_{\mathrm{SEIR}}(\tau)$: for example, setting $a_i(\tau) \equiv A_{\mathrm{SEIR}}(\tau)$ for all individuals gives a "smooth" profile in which each person's infectiousness follows the population average (**Figure XX**). At the opposite extreme, $a_i(\tau) = (\beta_{\mathrm{SEIR}}/\alpha_{\mathrm{SEIR}})\delta_{\tau_i^*}(\tau)$, with $\delta_{\tau_i^*}(\tau)$ a delta function that has mass at τ_i^* drawn from $g(\tau) = A_{\mathrm{SEIR}}(\tau)/R_0$, gives a "spike" profile in which each person's infectiousness is concentrated at a single moment (**Figure XX**). All three families — stepwise, smooth, and spike — produce the same $A_{\mathrm{SEIR}}(\tau)$ and hence the same deterministic, mean-field epidemic dynamics, but the resulting stochastic dynamics may differ.

While estimates of R_0 and $g(\tau)$, and thus $A(\tau)$, are commonplace for most pathogens, we often have little understanding of $a_i(\tau)$'s shape. Viral kinetics offer an indirect window into the individual-level timing and intensity of infectiousness, but the precise mapping is poorly understood. Contact tracing data are often used to inform ν_i, but capturing $g_i(\tau)$ from the same data is challenging due to uncertainty in infection onset times and imperfect case ascertainment. Nevertheless, we can reason about the potential shapes of $a_i(\tau)$: the full variability in $g(\tau)$ must arise from both within-individual and between-individual variation in infection timing, suggesting that $a_i(\tau)$ is generally narrower than $A(\tau)$. Some evidence indicates that, for SARS-CoV-2 and influenza, the window of peak transmissibility at the individual level may be extremely short — even less than a day — which is substantially narrower than the pathogens' overall generation interval distributions [Goyal et al., 2021]. So, while estimates of $a_i(\tau)$ are lacking, the available evidence suggests that the shape of $a_i(\tau)$ may differ meaningfully from that of $A(\tau)$.

However, a central question remains: to what extent does the shape of $a_i(\tau)$, holding $A(\tau)$ fixed, impact epidemic dynamics? Several lines of evidence suggest it could matter substantially, but the question has not been studied systematically. Within the SEIR framework (which assumes a stepwise individual infectiousness profile), replacing exponentially-distributed latent and infectious periods with more realistic distributions can dramatically alter epidemic trajectories, changing peak size, epidemic duration, growth rate, outbreak probability, and estimates of R_0, even when the mean periods are held fixed [Lloyd, 2001, Wearing et al., 2005, Britton and Lindenstrand, 2009]. Replacing the stepwise $a_i(\tau)$ with a time-varying, viral kinetics-informed infectiousness profile can generate significant variability in epidemic peak size and timing [Li et al., 2023]. In applied settings, the shape of $a_i(\tau)$ enters implicitly: models of contact tracing effectiveness depend on "the course of individual infectivity" as a key input [Klinkenberg et al., 2006], and models of testing-based interventions rely on assumptions about the connection between viral load and infectiousness, which generally produce individual infectiousness profiles that are substantially narrower than the population average [Larremore et al., 2021]. Yet in each of these cases, the shape of $a_i(\tau)$ is either a modeling assumption adopted for a different purpose or a feature varied at the population level; no study has systematically isolated the impact of variation in $g_i(\tau)$ on epidemic dynamics while holding R_0, $g(\tau)$, and $A(\tau)$ constant at the population level.

Here, we address this gap by conducting a systematic analysis of how the shape of $a_i(\tau)$ (*i.e.*, the decomposition of $g(\tau)$'s variability into within- *vs.* between-individual components) affects epidemic behavior, holding all population-level quantities constant. To do so, we introduce a flexible modeling framework, parameterized by a smoothness parameter $\psi \in [0, 1]$, that interpolates between the "spike" and "smooth" extremes. This approach extends the renewal equation framework to accommodate individual-level variation in infectiousness timing [Breda et al., 2012]. We assess how the shape of $a_i(\tau)$, independent of other factors, creates variation in epidemic establishment times, impacts the inference of the epidemic growth rate r and the generation interval distribution $g(\tau)$, and introduces additional overdispersion in secondary infection counts. We also determine how $a_i(\tau)$'s shape modulates the effectiveness of detect-and-isolate interventions and gathering size restrictions. We illustrate these findings using three respiratory pathogens — influenza, SARS-CoV-2 omicron, and measles — but emphasize that the methods and findings are general. We also address the question of whether the shape of $a_i(\tau)$ can be inferred from contact-tracing data, and conclude with a discussion of implications for epidemiological modeling and outbreak control.

2 Results

2.1 *A time-shifted Gamma model flexibly captures individual-level variation in infectiousness timing*

We begin by introducing a decomposition of $g(\tau)$ into within- and between-individual components. We seek an analytically tractable definition of $g_i(\tau)$ such that $g(\tau) = E_i[g_i(\tau)]$, and where the width of $g_i(\tau)$ is governed by a single smoothness parameter $\psi \in [0,1]$ that interpolates between the "spike" and "smooth" extremes (**Figure XX**).

To achieve this, we first assume that $g(\tau)$ follows a Gamma(α, β) distribution with rate $\beta = \alpha/\bar{g}$, where $\bar{g}$ is the mean generation time. The Gamma distribution is a natural and common choice for modeling generation interval distributions. Next, we define each person's individual intrinsic generation interval distribution as

$$g_i(\tau) = \begin{cases} 0 & \tau \leq l_i \\ f_\epsilon(\tau - l_i) & \tau > l_i \end{cases} \tag{3}$$

where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ is a random latent period and f_ϵ is the Gamma$(\psi\alpha, \beta)$ density (**Figure XX**). Equivalently, in terms of random variables, the secondary infections j from an index case i are distributed at times $\tau_{ij} = l_i + \epsilon_j$, where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ and $\epsilon_j \overset{iid}{\sim} \text{Gamma}(\psi\alpha, \beta)$.

Since l_i and ϵ_j are independent Gamma-distributed random variables with common rate β, their sum follows a $\text{Gamma}((1-\psi)\alpha + \psi\alpha, \beta) = \text{Gamma}(\alpha, \beta)$ distribution; thus $E_i[g_i(\tau)] = g(\tau)$, as required. Furthermore, when $\psi \to 1$, the latent period vanishes ($l_i \to 0$) and f_ϵ approaches the full Gamma(α, β) density, which gives the "smooth" extreme (no between-individual variation). Similarly, when $\psi \to 0$, f_ϵ concentrates to a point mass and $g_i(\tau)$ becomes a delta function at $l_i \sim \text{Gamma}(\alpha, \beta)$, which gives the "spike" extreme (no within-individual variation). We generally assume that ψ is a single value shared by all individuals for a given pathogen, but the framework is more general: the Gamma addition property ensures that drawing individual ψ_i from a distribution on $[0,1]$ still yields $E_i[g_i(\tau)] = g(\tau)$; thus, it is possible to model scenarios where individuals have a mixture of narrow and broad individaul infectiousness profiles.

2.2 *Expressing the individual reproduction number in terms of* $g_i(\tau)$

The individual reproduction number, ν_i, is the expected number of secondary infections that an index case i is expected to produce in an otherwise susceptible population; it is the individual-level analog of R_0. Intuitively, ν_i depends on a person's baseline "biological" infectiousness (*e.g.*, the amount of pathogen shed over the course of infection) and on the overlap between their infectiousness timing and their contact patterns. Formally,

$$\nu_i = b_i \int_0^\infty g_i(\tau) c_i(t_i + \tau) d\tau \tag{4}$$

Here, b_i is the "biological infectiousness", defined as the expected number of secondary infections person i would produce under a time-uniform reference contact process, which we denote $\bar{c} \equiv 1$. To account for time-varying contacts, we introduce a contact proces $\mathbf{c}(t)$: a nonnegative, wide-sense stationary stochastic process with time-mean $E_t[\mathbf{c}(t)] = 1$. Each person's contacts follow a draw from this stochastic process, $c_i(t)$, representing the proportional scaling of that person's contact rate relative to the reference level (1) at calendar time t. The constraint $E_t[\mathbf{c}(t)]$ is an ensemble average over all possible realizations at time t; individual realizations c_i need not have time-average 1, allowing for individuals with persistently higher or lower contact rates than the population average.

For most of the analyses that follow, we fix $c(t) \equiv \bar{c} = 1$ in an attempt to vary as few parameters as possible. However, the full expression in Eq. **??** is useful for considering how the shape of $g_i(\tau)$ interacts with variable contact patterns.

2.3 Punctuated infectiousness profiles generate more variably-timed epidemics

Early-outbreak stochasticity can impact when an epidemic becomes established, which in turn shifts the epidemic's full trajectory [Morris et al., 2024]. Formally, the cumulative number of infections $Z(t)$ in the early stage of an epidemic can be described as a branching process

$$Z(t) = W(t)e^{rt} \tag{5}$$

where $W(t)$ is a stationary stochastic process with initial condition $W(0)$ and growth rate r. The process $W(t)$ converges almost surely to a random variable W as $t \to \infty$, which can be interpreted as a random initial condition on the branching process. Large values of W correspond to random successful early transmission events, inflating the effective initial condition. With a change of variables, this random initial condition can be expressed as a random time shift ς, where

$$\varsigma = \frac{\log W - \log E[W]}{r} \tag{6}$$

and the branching process can be expressed as

$$Z(t) \approx e^{r(t+\varsigma)} \tag{7}$$

Thus, W, and ς by extension, provide a window into the long-term impact of early-outbreak stochasticity [Morris et al., 2024].

Narrow individual infectiousness profiles ($\psi \to 0$) inflate $\mathrm{Var}[W]$ relative to smooth profiles ($\psi \to 1$), which makes epidemic timing more variable (increases $\mathrm{Var}[\varsigma]$) and shifts the typical epidemic slightly later ($E[\varsigma]$ more negative). Analytic results confirm that

$$\mathrm{Var}[W] \propto c(R_0, \alpha)^{(1-\psi)\alpha} \tag{8}$$

Here, $c(R_0, \alpha)$ is a constant that depends on R_0 and α; it is monotonically increasing in R_0 and is greater than 1 when $R_0 > 1$. Thus, $\mathrm{Var}[W]$ is largest when R_0 is high and ψ is small (*i.e.*, $g_i(\tau)$ is narrow), with the impact of ψ modulated by α (the shape/skewness of $g(\tau)$). If we assume, following Morris et al. [2024], that the distribution of $W|\mathrm{surv}$ (*i.e.*, conditioning on epidemic non-extinction) is well-approximated by a Gamma distribution, the inflated $\mathrm{Var}[W]$ translates into higher $\mathrm{Var}[\varsigma|\mathrm{surv}]$ and more negative $E[\varsigma|\mathrm{surv}]$ (**Supplementary Methods**).

Simulations indicate that these ψ-driven changes in the distribution of W and ς yield meaningful differences in epidemic trajectories (**Figure XX**). We simulated 5,000 epidemics in a population of size 10,000 for each of three benchmark pathogens: influenza, SARS-CoV-2 omicron, and measles, with R_0 and $g(\tau)$ informed by literature values (**Table XX**). We varied $\psi \in \{0, 0.5, 1\}$ and measured the time to reach a cumulative prevalence of 5% (500 total infections), which we used as a marker of epidemic establishment. By day 20, only 55% of simulated SARS-CoV-2 epidemics with $\psi = 0$ had reached 5% prevalence, while 82% of epidemics with $\psi = 1$ had reached the same threshold (**FigureXX**). Similarly, by day 33, just 58% of simulated measles epidemics with $\psi = 0$ had reached 5% prevalence, while 85% of epidemics with $\psi = 1$ had reached the same threshold (**Figure XX**). These differences reflect the higher likelihood of late-shifted epidemics for when the individual infectiousness profile is narrow ($\psi \to 0$). The effect is strongly modulated by R_0: for influenza, with the lowest R_0 of 2, differences in epidemic timing across ψ-values were hardly discernible (**Figure XX**).

2.4 *Punctuated infectiousness profiles make estimates of r and $g(\tau)$ less certain*

Inferring the growth rate r and the generation interval distribution $g(\tau)$ are critical tasks during an emerging epidemic and are key ingredients for calculating R_0. Punctuated individual infectiousness profiles make both tasks more challenging. First, punctuated infectiousness profiles cause daily case counts to be more variable (overdispersed relative to the mean), translating into greater uncertainty in r. Assuming Poisson(R_0)-distributed secondary infection counts, we can calculate the index of dispersion of $N(\tau)$, the number of infections generated by a single person falling on the day $[\tau, \tau+1)$ after infection:

$$\mathrm{ID}[N(\tau)] \equiv \frac{\mathrm{Var}[N(\tau)]}{E[N(\tau)]} = 1 + R_0 \frac{\mathrm{Var}[Q(\tau)]}{E[Q(\tau)]} \tag{9}$$

Here, $Q(\tau) = \int_\tau^{\tau+1} f_\epsilon(u-l)du$ is the probability that a secondary infection lands on day $[\tau, \tau+1)$, with $l \sim \mathrm{Gamma}((1-\psi)\alpha, \beta)$. In the smooth extreme ($\psi = 1$), $l_i = 0$ and $f_\epsilon = g$, so $Q(\tau)$ is deterministic: thus, $\mathrm{Var}[Q(\tau)] = 0$ and $\mathrm{ID}[N(\tau)] = 1$. In the spike extreme ($\psi = 0$), f_ϵ is a point mass, so $Q(\tau)$ is Bernoulli, equalling 1 with probability $q(\tau) \equiv \int_\tau^{\tau+1} g(u)du$ (*i.e.*, the probability that l falls in $[\tau, \tau+1)$) and 0 otherwise; thus, $\mathrm{Var}[Q_i(\tau)] = q(\tau)(1-q(\tau))$ and $\mathrm{ID}[N(\tau)] = 1 + R_0(1-q(\tau)) > 1$. Thus, narrower infectiousness profiles increase overdispersion in daily case counts, with the effect size modulated by R_0. Since infectors contribute independently to the population-wide daily case count, this overdispersion carries through to the aggregate daily case count and translates into increased uncertainty in the estimate for r.

Simulations illustrate the additional uncertainty in estimates of r that arises from narrow individual infectiousness profiles. We simulated 5,000 epidemics in infinite populations (to avoid depletion-of-susceptibles effects) for each of the three benchmark pathogens (influenza, SARS-CoV-2 omicron, measles). We ran each simulation for 7 days after reaching 100 cumulative cases, then binned infections into daily counts. To estimate growth rate, we fit a Poisson generalized linear model to the daily infection counts []. Punctuated individual infectiousness profiles systematically increased variability in the estimates of r: across the simulations, the standard deviation in $\hat{r}_{\mathrm{MLE}}$ (the maximum-likelihood estimate of r) was between xx and xx times higher for $\psi = 0$ than for $\psi = 1$, with the largest effect for measles, which has the highest R_0 (12) (**Figure XX**).

Punctuated individual infectiousness profiles can also make estimates of $g(\tau)$ more uncertain. Typically, $g(\tau)$ is estimated using contact tracing data capturing serial intervals, from which generation intervals are inferred by assuming an observation delay model. For n observed serial intervals, Kish's design effect for cluster sampling [] gives an effective sample size of $n_{\mathrm{eff}} = n/(1 + (R_0 - 1)\,\mathrm{ICC})$, where ICC is the intraclass correlation coefficient. Under the time-shifted Gamma model, and assuming an independent $\mathrm{Gamma}(a_{\mathrm{obs}}, b_{\mathrm{obs}})$ observation delay applied to each infection time, the intraclass correlation coefficient is (**Supplementary Materials**)

$$\mathrm{ICC} = \frac{(1-\psi)\alpha/\beta^2 + (a_{\mathrm{obs}}/b_{\mathrm{obs}}^2)^2}{\alpha/\beta^2 + 2(a_{\mathrm{obs}}/b_{\mathrm{obs}}^2)^2} \tag{10}$$

Thus, n_{eff} is minimized when $\psi \to 0$, with the size of the ψ-effect modulated by the variance of $g(\tau)$ (*i.e.*, α/β^2).

Simulations demonstrate that, if within-cluster correlations are ignored and serial intervals are instead treated as noisy *iid* draws from $g(\tau)$, punctuated individual infectiousness profiles cause posterior estimates of α and β to frequently miss the true value (**Figure XX**). We simulated the offspring of 100 index cases, introduced Gamma-distributed observation noise with mean and variance of 4 days ($a_{obs} = 4$, $b_{obs} = 1$), and extracted the serial intervals. We then used these serial intervals to estimate posterior distributions for α and β, treating the observations as independent. We repeated this process 500 times for each of the three benchmark pathogens. The posterior coverage, defined as when 95% credible interval contained the true value, decreased dramatically as the individual infectiousness profile narrowed. Across pathogens, coverage ranged from XX–XX for α and XX–XX for β when $\psi = 1$. This dropped to XX–XX for α and XX–XX for β when $\psi = 0$. This reduced coverage is a direct consequence of treating the serial intervals as overly informative about $g(\tau)$, whereas when $\psi \to 0$, the offspring of an index case reflect effectively just one draw from $g(\tau)$.

2.5 *Punctuated infectiousness profiles interact with time-varying contacts produce superspreading*

Superspreading arises when secondary infection counts are overdispersed, *i.e.*, when the variance in the number of secondary infections produced by an individual exceeds the mean. Formally, if the number secondary infections χ_i produced by person i are Poisson(ν_i)-distributed, overdispersion arises when ν_i varies across individuals. This overdispersion can powerfully impact epidemic dynamics, making epidemics less likely to establish but potentially more explosive when they do [Lloyd-Smith et al., 2005].

From Eq. **??**, we see that variation in ν_i can arise in three fundamental ways: variation in biological infectiousness, b_i; variation in individual contact levels, $E_t[c_i(t)]$; and the variable interaction between infectiousness timing and the contact function, $g_i(\tau)c_i(t_i + \tau)$. The first two contributors to overdispersion (so-called "super-shedders" and "super-contactors") have been studied extensively; here, we examine the impact of the g_i–c_i interaction as a third mechanism of superspreading, holding $b_i \equiv R_0$ and $E_t[c_i(t)] = 1$.

We considered two canonical contact processes: a "periodic" contact model, where $c_i(t) = 1 - c_{amp}\cos\left(2\pi t/c_{per}\right)$, and a "stochastic" contact model, where $c_i(t)$ switches to a new Gamma(σ, σ)-distributed level at arrivals of a Poisson process with rate λ (**Figure XX, Methods**). These models represent two axes of contact variation: the periodic model has variation over time, which carries through to the population level ($E_i[c_i(t)]$ is periodic in time), but no variation across individuals. Alternatively, the stochastic model varies across individuals, but the population-average contact process is identically equal to 1 ($E_i[c_i(t)] = 1$). (Note: the wide-sense stationarity of the periodic contact model is achieved by considering a random infection time t_i which is uniformly distributed over one period; thus, the process is stationary with respect to infection times, but not with respect to calendar time).

Following Lloyd-Smith et al. [2005], we quantify overdispersion in terms of the parameter k, where the distribution of secondary infections is approximated by $\chi_i \sim \text{NegBin}(R_0, k)$ (exact when ν_i are Gamma-distributed). Here, $E[\chi_i] = R_0$ and $\text{Var}[\chi_i] = R_0(1 + R_0/k)$; thus, $k \in (0, \infty)$ captures overdispersion, with $k \to \infty$ yielding Poisson-distributed χ_i and $k \to 0$ yielding greater overdispersion. For convenience, we introduce the parameter $\text{OD} = 1/k$, so that large OD corresponds to high overdispersion.

For the periodic contact model, we can derive a simple closed-form expression for OD in terms of the model parameters (**Supplementary Methods**):

$$\text{OD} = \frac{c_{amp}^2}{2}\left(1+\left(\frac{2\pi\bar{g}}{\alpha c_{per}}\right)^2\right)^{-\psi\alpha} \tag{11}$$

From this expression, we see that $c_{amp} \to 0$ yields OD $\to 0$, *i.e.*, when contacts do not vary over time, we recover the standard Poisson-distributed secondary infection distribution with no overdispersion. When $c_{amp} \neq 0$, narrower individual infectiousness profiles ($\psi \to 0$) increase OD; and the sensitivity of OD to ψ depends on the magnitude of $\eta = 2\pi\bar{g}/(\alpha c_{per})$. When $\eta \gg 1$, which happens for example when $c_{per} \ll \bar{g}$, OD is small regardless of ψ. In this case, the contact process varies so quickly that virtually all infectiousness profiles average over the entire contact process, yielding little overdispersion. Alternatively, when $\eta \ll 1$, which can happen when $c_{per} \gg \bar{g}$, OD $\approx c_{amp}^2/2$ regardless of ψ. In this case, the contact process varies so slowly that all infectiousness profiles, whether broad or narrow, "see" only a small part of the contact process' variation, causing maximal overdispersion that is insensitive to ψ. The impact of ψ is thus most visible when $\eta \approx 1$, in which case $\psi \to 1$ causes the infectiousness profile to effectively average over the contact variation, while $\psi \to 0$ translates the contact variation into meaningful overdispersion. For the three benchmark pathogens, $\bar{g}$ ranges from 3.2 days to 12.2 days. Thus, weekly variation in contacts ($c_{per} = 7$) causes OD to vary meaningfully with ψ. Daily variation ($c_{per} = 1$) gets averaged out, yielding little overdispersion unless $\psi \approx 0$. Monthly variation ($c_{per} = 30$) leads to OD $\approx c_{amp}^2/2$ regardless of ψ. While the analytic form is less revealing, similar patterns hold for the stochastic contact model, with the switching rate λ and dispersion parameter σ playing analogous roles to c_{per} and c_{amp}.

Simulations illustrate the impact of ψ on overdispersion and on outbreak survival probabilities. To capture the relationship between contact variability (c_{amp} or σ), ψ, and OD, we simulated the offspring of 10,000 index cases for each of the three benchmark pathogens and both of the contact processes across a range of ψ, c_{amp}, and σ values (**Figure XX**). We estimated OD by moment-matching the mean and variance of the offspring counts to a Negative Binomial distribution. Across pathogens, overdispersion was highest for narrow infectiousness profiles ($\psi \to 0$) and high contact variability (large c_{amp}, small σ) (**Figure XX**). Furthermore, we simulated 1,000 epidemics in a population of size 10,000 for each of the three benchmark pathogens and for both of the contact process, and then calculated the fraction of outbreaks that failed to surpass a cumulative incidence of 5% (500 total infections). Extinction probabilities were systematically higher for more punctuated individual infectiousness profiles (**Table XX**), with the greatest discrepancies in measles (?), for which the $\psi = 0$ extinction probabilities were XX (periodic) and XX (stochastic) *vs.* XX (periodic) and XX (stochastic) for $\psi = 1$.

2.6 Detect-and-isolate interventions are sensitive to individual infectiousness timing

Detect-and-isolate (D&I) interventions are critical non-pharmaceutical interventions, aimed at identifying individuals at an early stage of infection to prevent onward spread. Detection can happen *via* the development of symptoms and/or diagnostic testing; thus, the timing of symptom onset and the window of test-based detectability, relative to the window of infectiousness, are critical factors influencing the success of D&I interventions [Fraser].

The efficacy of D&I interventions can be summarized in terms of testing effectiveness (TE), defined as the expected proportion by which a D&I intervention reduces population-level transmission [Middleton and Larremore, 2024]. Formally,

$$\text{TE} = \rho\eta P(\tau > D) \tag{12}$$

where $\rho \in [0, 1]$ is the participation rate, $\eta \in [0, 1]$ is the effectiveness of isolation (so that post-detection transmission attempts succeed with probability $1 - \eta$), τ is the time of a transmission attempt relative to the index case's infection time, and D is the detection time. When D is anchored to the index case's infection time (*e.g.*, the detectability window spans 2–7 days after infection), TE depends only on the shape of the generation interval distribution; the shape of the individual infectiousness profile integrates out (**Supplementary Methods**). If, however, D is anchored to the individual infectiousness profile (*e.g.*, the window of detectability extends $\pm$ 3 days from peak infectiousness), TE depends on the width of the individual infectiousness profile, as we illustrate here. Anchoring D to the infectiousness profile is arguably the more realistic scenario since symptom onset, test-based detectability, and peak pathogen shedding are all linked to viral load, which in turn shapes the individual's infectiousness trajectory [Boyer].

Narrow infectiousness profiles make TE sensitive to small shifts in detection time. To illustrate this, we considered two detection mechanisms: symptom-based isolation, where symptoms arise at a Normal$(\delta_{symp}, \sigma_{symp})$-distributed offset from the peak infectiousness and isolation is immediate; and test-based screening, where diagnostic tests are administered at fixed intervals of length Δ_{test}, and the first test that falls within a detectability window spanning δ_{pre} through δ_{post} days before/after peak infectiousness triggers isolation after a (λ_{act})-distributed turnaround delay. TE can then be calclulated *via* numerical integration (**Methods**). For both testing scenarios, TE dropped as the typical detection time shifted from before to after peak infectiousness, with the sharpest drops corresponding to the narrowest individual infectiousness profiles (**Figure XX**). When the typical detection time was near the time of peak infectiousness, this led to substantial differences in TE across ψ-values; for symptom-based screening, the absolute discrepancy reached as much as xx% (TE = XX for $\psi = 1$, TE = XX for $\psi = 0$) for measles(?) when $\delta_{symp} = XX$ and $\sigma_{symp} = XX$.

Beyond TE, D&I interventions produce two additional ψ-dpeendent effects: generation inteval truncation, which can accelerate epidemic growth; and increased overdispersion in secondary infection counts, which increases the probability of epidemic extinction. Under a D&I intervention, the effective generation interval distribution becomes (**Supplementary Methods**)

$$g^*(\tau) = \frac{(1 - \eta\sigma)g(\tau) + \eta\sigma \int_0^{\min(\tau, D)} f_I(\tau - \epsilon) f_\epsilon(\epsilon)\, d\epsilon}{1 - \mathrm{TE}} \tag{13}$$

For maximally punctuated individual infectiousness profiles, f_ϵ is a delta function, yielding $g^*(\tau) = g(\tau)$; that is,

2.7 *Gathering size restrictions are sensitive to individual infectiousness timing*

2.8 *Identifiability of the individual infectiousness profile's shape depends on R_0 and detection delays*

2.9 *Estimates of individual infectiousness timing for various pathogens*

3 Discussion

- For the ς section, the shift in $E[\varsigma]$ is minor; the inflation of $\text{Var}[\varsigma]$ is the real story.
- For the sensitivity of TE: note that shifts in symptom timing are commonplace (happend with SARS-CoV-2), and small changes in testing frequency are easy operational decisions.
- Flesh out the counter-intuitive impact of TE/GE contraction/overdispersion for detect-and-isolate, and Reff/overdispersion for gathering size restrictions.

4 Methods

5 Modeling framework

5.1 *Accounting for within- vs. between-individual variation in infectiousness timing: the time-shifted Gamma model*

We begin by introducing a modeling framework that decomposes $g(\tau)$'s variance into within- and between-individual components *via* a single parameter. We seek an analytically tractable definition of $g_i(\tau)$ such that $g(\tau) = E_i[g_i(\tau)]$, and where the width of $g_i(\tau)$ is governed by a smoothness parameter $\psi \in [0, 1]$ that interpolates between the "spike" and "smooth" extremes.

To achieve this, we first assume that $g(\tau)$ follows a Gamma(α, β) distribution with rate $\beta = \alpha/\bar{g}$, where $\bar{g}$ is the mean generation time. The Gamma distribution is a natural and common choice for modeling generation interval distributions. Next, we define each person's individual intrinsic generation interval distribution as

$$g_i(\tau) = \begin{cases} 0 & \tau \leq l_i \\ f_\epsilon(\tau - l_i) & \tau > l_i \end{cases} \tag{14}$$

where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ is a random latent period and f_ϵ is the Gamma$(\psi\alpha, \beta)$ density (**Figure XX**). Equivalently, in terms of random variables, the secondary infections j from an index case i are distributed at times $\tau_{ij} = l_i + \epsilon_j$, where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ and $\epsilon_j \overset{iid}{\sim} \text{Gamma}(\psi\alpha, \beta)$.

Since l_i and ϵ_j are independent Gamma-distributed random variables with common rate β, their sum follows a Gamma$((1-\psi)\alpha + \psi\alpha, \beta) =$ Gamma(α, β) distribution; thus $E_i[g_i(\tau)] = g(\tau)$, as required. Furthermore, when $\psi \to 1$, the latent period vanishes ($l_i \to 0$) and f_ϵ approaches the full Gamma(α, β) density, which gives the "smooth" extreme (no between-individual variation). Similarly, when $\psi \to 0$, f_ϵ concentrates to a point mass and $g_i(\tau)$ becomes a delta function at $l_i \sim \text{Gamma}(\alpha, \beta)$, which gives the "spike" extreme (no within-individual variation).

5.2 *Accounting for time-varying contacts*

For $A(\tau)$ and $a_i(\tau)$ to be functions of τ (time since infection) alone, and not also of calendar time t, it is necessary to assume that interpersonal contact rates do not vary over time. We follow this convention for most of our analyses. However, the interaction between punctuated $g_i(\tau)$ and time-varying contacts may be consequential. Intuitively, if a person's infectiousness is concentrated in a narrow window, then their realized transmission depends heavily on their contact level during that window; whereas if their infectiousness is spread over a long period, contact fluctuations will get averaged out. Thus, we briefly introduce a generalization that accounts for time-varying contacts. We also introduce two specific contact models that serve as test beds.

Under time-uniform contacts, $a_i(\tau) = \nu_i g_i(\tau)$, where ν_i and $g_i(\tau)$ are defined with respect to a reference contact level which we denote $\bar{c} \equiv 1$. To allow for time-varying contacts, we introduce a contact process $\mathbf{c}(t)$: a nonnegative, wide-sense stationary stochastic process with time-mean $E_t[\mathbf{c}(t)] = 1$. Each person i has a contact modulator $c_i(t)$ that is a realization of $c(t)$, representing the proportional scaling of that person's contact rate relative to the reference level at calendar time t. The constraint $E_t[\mathbf{c}(t)] = 1$ is an ensemble average over all possible realizations at time t; individual realizations c_i need not have time-average 1, allowing for individuals with persistently higher or lower contact rates than the population average. We additionally define the "biological infectiousness", b_i, as the expected number of secondary infections person i would produce under $\bar{c} \equiv 1$. The individual infectiousness profile thus generalizes to

$$a_i(t_i, \tau) = b_i g_i(\tau) c_i(t_i + \tau) \tag{15}$$

where t_i is person i's infection time. If $c(t) \equiv \bar{c} = 1$, this reduces back to the familiar $a_i(\tau) = \nu_i g_i(\tau)$, with $b_i = \nu_i$. More generally,

$$\nu_i = b_i \int_0^\infty g_i(\tau) c_i(t_i + \tau)\, d\tau \tag{16}$$

i.e., the individual reproduction number is a function of the person's inherent biological infectiousness b_i as well as the interaction between their infectiousness timing and their contact process. Likewise, the realized generation interval distribution is

$$g_i^{\text{re}}(t_i, \tau) = \frac{g_i(\tau) c_i(t_i + \tau)}{\int_0^\infty g_i(u) c_i(t_i + u)\, du} \tag{17}$$

So, an equivalent rendering of Eq. 14 is

$$a_i(t_i, \tau) = \nu_i g_i^{\text{re}}(t_i, \tau) \tag{18}$$

For the population-level quantities, it can be shown that

$$R_0 = E_i[\nu_i] = E_i[b_i] \qquad \text{and} \qquad g(\tau) = E_i[b_i g_i(\tau)]/R_0 \tag{19}$$

where the second equality follows from the assumed independence of (b_i, g_i) and c_i and the constraint $E[\mathbf{c}(t)] = 1$ (**Supplementary Methods**). This gives the population-level infectiousness profile

$$A(t, \tau) = R_0 g(\tau) C(t + \tau) \tag{20}$$

where $C(t) = E_i[c_i(t)]$, the population-average contact level at calendar time t.

As test cases, we consider two models. We define the "periodic" contact model

$$c_i(t) = 1 - c_{amp} \cos\left(\frac{2\pi t}{c_{per}}\right) \quad \forall i \tag{21}$$

with amplitude $c_{amp} \in [0, 1]$ and period $c_{per} > 0$ (**Figure XX**). In this model, all individuals share the same deterministic contact function; this is a degenerate case of the contact process $\mathbf{c}(t)$ in which every realization is identical. While there is temporal variation within each person's contacts, there is no variation across individuals. The process' stationarity is achieved by considering a random infection time t_i which is uniformly distributed over one period (which may approximately hold during the period of exponential epidemic growth) [Papoulis, 1965]. At the population level, $C(t) = E_i[c_i(t)] = 1 - c_{amp} \cos(2\pi t / c_{per})$.

We also define the "stochastic" contact model, where each person's $c_i(t)$ is a piecewise-constant process that switches to a new level at arrivals of a Poisson process with rate λ (**Figure XX**). The contact levels are drawn independently from a Gamma(σ, σ) distribution, which has mean 1 (satisfying the constraint $E_t[\mathbf{c}(t)] = 1$) and variance $1/\sigma$. This creates a two-parameter family indexed by the dispersion parameter $\sigma > 0$ and switching rate λ. Smaller σ yields more heterogeneous contacts, while large σ yields a bell-shaped distribution concentrated around 1. As $\sigma \to \infty$, the distribution collapses to a point mass at 1, recovering the constant-contact reference process. In contrast to the periodic model, each person's contact trajectory is unique, but the population average is constant in time: $C(t) = E_i[c_i(t)] = 1$.

5.3 *Simulation approach*

To investigate the impact of individual-level infectiousness timing on epidemic dynamics, we use both closed-form analytic results and stochastic epidemic simulations. We initialized simulations with a single infectious individual in a finite population of $N = 10,000$. For each simulation, we drew a latent period l_i from a Gamma$((1-\psi)\alpha, \beta)$ distribution and, where relevant, a contact trajectory $c_i(t)$ for each person i. Upon infection, the number of secondary infection attempts χ_i was drawn from a Poisson(b_i) distribution. Throughout, we assume $b_i \equiv R_0$, *i.e.*, no interpersonal variation in biological infectiousness, to isolate the effect of infectiousness timing from that of variation in the individual reproduction number. For each infection attempt j, a random time offset ϵ_j was drawn from a Gamma$(\psi\alpha, \beta)$ distribution, so that the timing of the attempt was $\tau_{ij} = l_i + \epsilon_j$. To account for time-varying contacts, we applied Poisson thinning, accepting each attempt with probability proportional to $c_i(t_i + \tau)$. Each attempted infection was then paired with a uniformly chosen individual from the population: if the recipient was susceptible, infection occurred; otherwise, the attempt failed.

Full simulation details are provided in the **Supplementary Methods**. For illustration, we focused on three sets of epidemiological parameters reflecting published estimates of R_0 and $g(\tau)$ of influenza, SARS-CoV-2 omicron, and measles (**Table XX**).

6 The impact of punctuated infectiousness on uncontrolled epidemics

6.1 *Punctuated infectiousness profiles generate later and more variably-timed epidemics.*

Early-outbreak stochasticity can impact when an epidemic becomes established, which in turn shifts the epidemic's full trajectory [Morris et al., 2024]. Simulations demonstrate that narrower $a_i(\tau)$ (*i.e.*, $\psi \to 0$) cause the epidemic establishment time to become more variable and to shift modestly later in expectation. For 5,000 simulations across each of the three benchmark pathogens and for $\psi \in \{0, 0.5, 1\}$, we measured the time to reach a cumulative prevalence of 5% (500 total infections), which we used as a marker of epidemic establishment. For influenza, the mean [5% quantile, 95% quantile] time to reach 500 infections was 24.2 days [17.8, 33.8] for $\psi = 1$ *vs.* 24.5 days [17.2, 34.9] for $\psi = 0$; for omicron, 18.1 days [14.8, 22.9] for $\psi = 1$ *vs.* 20.3 days [13.4, 30.4] for $\psi = 0$; and for measles, 31.3 days [28.6, 34.6] for $\psi = 1$ *vs.* 32.5 days [26.4, 39.5] for $\psi = 0$. This translated into meaningful differences in epidemic timing, particularly for SARS-CoV-2 omicron and measles: on day 20, 82% of simulated SARS-CoV-2 omicron outbreaks with $\psi = 1$ had reached 500 cases, while only 55% of outbreaks with $\psi = 0$ had reached the same threshold (**Figure XX**). Similarly, by day 33, 84% of simulated measles outbreaks with $\psi = 1$ had reached 500 cases, while only 58% of outbreaks with $\psi = 0$ had reached the same threshold (**Figure XX**).

These trends can be understood intuitively by considering the two extremes of ψ. When infectiousness is maximally punctuated ($\psi = 0$), all of an index case's secondary infections occur at a single moment $\tau_{ij} l_i$, a single draw from $g(\tau)$. When infectiousness is maximally smooth ($\psi = 1$), the secondary infection times are independent draws from $g(\tau)$. For the smooth profile, the earliest secondary infection is the minimum order statistic of χ_i independent draws from $g(\tau)$, which falls earlier in expectation than a single draw. This order statistic effect is magnified as R_0, and thus the typical χ_i, increases. Since the speed at which an epidemic establishes depends disproportionately on the earliest transmissions in each generation, smooth profiles confer an effective head start. Conversely, under the punctuated profile, the entire next generation's timing is determined by a single random quantity l_i; thus, an unusually early or late draw propagates fully into the epidemic trajectory without being offset by more "typical" draws, and inflates the variability in epidemic establishment times.

Analytic results confirm these trends. Following Morris et al. [2024], we describe the cumulative number of infections $Z(t)$ at time t during the early-epidemic exponential growth phase as a rescaled branching process

$$W(t) = e^{-rt} Z(t) \tag{22}$$

with initial condition $W(0)$ and growth rate r. The process W converges almost surely to a random variable W as $t \to \infty$. W can be loosely interpreted as a random initial condition on the branching process that captures the effect of early-outbreak stochasticity: large values of W correspond to random successful early transmission events, inflating the effective initial condition. With a change of variables, this random initial condition can be expressed as a random time shift ς, where

$$\varsigma = \frac{\log W - \log E[W]}{r} \tag{23}$$

and the branching process can be expressed as

$$Z(t) \approx e^{r(t+\varsigma)} \tag{24}$$

For notational simplicity, we assume that epidemics are always seeded with a single infectious case; thus $E[W] = Z_0 = 1$ and $\varsigma = \log W/r$.

The characterization of W and its transformation into a time shift ς follow standard theory [Morris et al., 2024]. Building upon this foundation, we can derive $\mathrm{Var}[W]$ as an explicit function of ψ (**Supplementary Methods**). When $g(\tau)$ follows the Gamma time-shift model, the variance of the random initial condition is:

$$\mathrm{Var}[W] = \frac{\left(\frac{1+z}{1+2z}\right)^{\alpha} + \left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} - 1}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} := \frac{\sigma_1^{\alpha} + \sigma_2^{(1-\psi)\alpha} - 1}{1 - \sigma_1^{\alpha}} \tag{25}$$

where $z = R_0^{1/\alpha} - 1$ and $\sigma_c = ((1+z)^c)/(1+2z)$. Thus, $\mathrm{Var}[W]$ depends on R_0, α (the shape/skewness of $g(\tau)$), and ψ. The punctuation parameter ψ enters Eq. 24 only once, in the exponent of σ_2. Since $\sigma_2 > 1$ when $R_0 > 1$, $\mathrm{Var}[W]$ is monotonically decreasing in ψ, so that narrower individual infectiousness profiles ($\psi \to 0$) maximize the variance. Furthermore, σ_2 is monotonically increasing in R_0 (**Supplementary Methods**), so larger R_0 amplifies the sensitivity of $\mathrm{Var}[W]$ to ψ.

If we assume, following Morris et al. [2024], that the distribution of $W|\mathrm{surv}$ (*i.e.*, conditioning on epidemic non-extinction) is well-approximated by a moment-matched Gamma distribution, we can translate Eq. 24 into implications for the time shift ς. Specifically, since ς is the logarithm of an approximately Gamma-distributed random variable, we can express $E[\varsigma|\mathrm{surv}]$ and $\mathrm{Var}[\varsigma|\mathrm{surv}]$ in terms of the digamma and trigamma functions (**Supplementary Methods**). These expressions confirm that narrower infectiousness profiles, which inflate $\mathrm{Var}[W]$, also lead to more negative $E[\varsigma|\mathrm{surv}]$ (*i.e.*, cause the expected time shift to fall later) and higher $\mathrm{Var}[\varsigma|\mathrm{surv}]$.

These findings also do not require us to assume that $g(\tau)$ follows the Gamma time-shift model. For any probability densities f_l and f_ϵ that define an individual latent period and infectiousness kernel, such that $f_l(\tau) * f_\epsilon(\tau) = g(\tau)$ where $*$ denotes convolution, it can be shown that shifting variance from f_ϵ to f_l increases $\text{Var}[W]$, with the effect maximized by the spike profile and modulated by R_0 (**Supplementary Methods**). If we assume the distribution of W is well-approximated by a Gamma distribution, the effects of f_ϵ's variance on $E[\varsigma|\text{surv}]$ and $\text{Var}[\varsigma|\text{surv}]$ follow as before.

6.2 *Punctuated infectiousness profiles yield less certain estimates of the epidemic growth rate and generation interval distribution.*

During an emerging epidemic, inferring the growth rate and the generation interval distribution are among the first critical tasks. These are key ingredients for calculating the reproduction number, which determines the effort needed to control an outbreak. Simulations indicate that punctuated individual infectiousness profiles cause estimates of r to be less certain. For each of the three benchmark pathogens, we simulated 5,000 epidemics in infinite populations. We ran each simulation for 7 days after reaching 100 cumulative cases and binned the infections into daily counts. Then, to estimate the growth rate, we fit a Poisson generalized linear model using maximum likelihood to the daily case counts. For all three pathogens, punctuated infectiousness profiles yielded more uncertain growth rate estimtes, with the ψ-related uncertainty increasing in R_0 (**Figure XX**). For influenza, the maximum likelihood growth rate estimate had standard deviation XX for $\psi = 1$, XX for $\psi = 0.5$, and XX for $\psi = 0$; for SARS-CoV-2 omicron, the growth rate estimate had standard deviation XX for $\psi = 1$, XX for $\psi = 0.5$, and XX for $\psi = 0$; and for measles, the growth rate estimate had standard deviation XX for $\psi = 1$, XX for $\psi = 0.5$, and XX for $\psi = 0$.

This uncertainty in r follows from overdispersion in the daily case counts generated by more punctuated individual infectiousness profiles (**Figure XX**). To see why, consider person i, infected at time t_i, who generates $\chi_i \sim \text{Poisson}(R_0)$ total secondary infections. Let $N_i(\tau)$ be the number of these infections that fall on day-since-infection $[\tau, \tau+1)$, and let $Q_i(\tau)$ be the probability that a given secondary infection from person i lands on that day:

$$Q_i(\tau) = \int_\tau^{\tau+1} f_\epsilon(u - l_i)\, du \tag{26}$$

where l_i is person i's realized latent period. The expectation of $Q_i(\tau)$ over latent period draws is

$$E[Q_i(\tau)] = \int_\tau^{\tau+1} g(u)\, du \equiv q(\tau) \tag{27}$$

which depends only on the population-level generation interval distribution and is therefore independent of ψ. Conditional on $Q_i(\tau)$, the Poisson thinning property gives $N_i(\tau) \mid Q_i(\tau) \sim \text{Poisson}(R_0\, Q_i(\tau))$. Applying the law of total variance over $Q_i(\tau)$ yields the index of dispersion (**Supplementary Methods**):

$$\text{ID}[N_i(\tau)] = \frac{\text{Var}[N_i(\tau)]}{E[N_i(\tau)]} = 1 + R_0 \frac{\text{Var}[Q_i(\tau)]}{E[Q_i(\tau)]} \tag{28}$$

The entire ψ-dependence enters through $\text{Var}[Q_i(\tau)]$. In the smooth case ($\psi = 1$), $f_\epsilon = g$ and $l_i = 0$ for all individuals, so $Q_i(\tau) = q(\tau)$ is deterministic: $\text{Var}[Q_i(\tau)] = 0$ and $\text{ID} = 1$, its minimal value. In the spike case ($\psi = 0$), f_ϵ is a point mass, so $Q_i(\tau)$ is Bernoulli, equaling 1 with probability $q(\tau)$ and 0 otherwise: $\text{Var}[Q_i(\tau)] = q(\tau)(1 - q(\tau))$ and $\text{ID} = 1 + R_0(1 - q(\tau))$, its maximal value. Thus, narrower infectiousness profiles increase overdispersion in daily case counts, with the effect modulated by R_0. Since infectors contribute independently, this overdispersion carries through to the aggregate daily infection count, and translates into increased uncertainty in the growth rate estimate.

Simulations also indicate that punctuated individual infectiousness profiles cause estimates of $g(\tau)$ to be less certain. Typically, $g(\tau)$ is estimated using contact tracing data capturing serial intervals, from which generation intervals are inferred by assuming an observation delay model. The generation intervals are typically assumed to represent *iid* draws from the generation interval distribution, which becomes increasingly inaccurate for narrower individual infectiousness profiles. In the limit of $\psi \to 0$, all generation intervals from a given index case are identical. The effective sample size thus ranges from roughly KR_0 when $\psi = 1$, where K is the number of index cases, to K when $\psi = 0$. Observation delays can obscure the correlation between secondary infection times, making it easy to assume that variation in secondary infection times from a single infector are due to differences in infection timing, when in fact they are only due to differences in the observation model.

Simulations demonstrate the impact of this phenomenon. We simulated offspring from 5,000 index cases for each of $\psi \in \{0, 0.2, 0.5, 0.7, 1\}$ and for each of the three benchmark pathogens. We applied a $\text{Gamma}(a_{obs}, b_{obs})$ observation error independently to each infection time (both index and secondary cases) to generate a set of simulated serial intervals. We then use these serial intervals to calculate the estimated generation interval distribution parameters, assuming that the serial intervals were independent. This represents a sort of statistically "best possible" scenario: we assumed ascertainment was perfect (all secondary cases from a given index cases were identified, and it was known which case was the index case and which were the secondary cases, even if a serial interval was negative); and we assumed perfect knowledge of the delay distribution parameters. Even so, we identified meaningful differences in the estimates of the generation interval distribution parameters (α, β) across ψ-values.

[[results]]

6.3 Punctuated infectiousness profiles interact with time-varying contact patterns to generate overdispersion in secondary infection counts

Superspreading arises when secondary infection counts are overdispersed, *i.e.*, when the variance in the number of secondary infections produced by an individual exceeds the mean. This overdispersion can powerfully impact epidemic dynamics, making epidemics less likely to establish but potentially more explosive when they do [Lloyd-Smith et al., 2005].

Mathematically, overdispersion in secondary infection counts is captured through variation in the individual reproduction number, ν_i. Secondary infection counts are typically assumed to follow a Poisson distribution: $\chi_i \sim \text{Poisson}(\nu_i)$. When $\nu_i \equiv R_0$ (no variation across individuals), $\text{Var}[\chi_i] = E[\chi_i] = R_0$, *i.e.*, there is no overdispersion. When ν_i itself is variable, this inflates the variance of χ_i relative to the mean, creating overdispersion. In the overdispersed scenario, secondary infection counts are often modeled using a Negative Binomial distribution:

$$\chi_i \sim \text{NegBin}(R_0, k) \tag{29}$$

with mean $E[\chi_i] = R_0$ and variance $\text{Var}[\chi_i] = R_0(1 + R_0/k)$. Thus, $k \in (0, \infty)$ captures overdispersion, with $k \to \infty$ yielding Poisson-distributed χ_i and $k \to 0$ yielding greater overdispersion. For convenience, we introduce the parameter $\text{OD} = 1/k$, so that large OD corresponds to high overdispersion.

From Eq. 15, we see that variation in ν_i can arise in three fundamental ways: variation in biological infectiousness, b_i; variation in individual contact levels, $E_t[c_i(t)]$; and the variable interaction between infectiousness timing and the contact function, $g_i(\tau)c_i(t_i + \tau)$. The first two contributors to overdispersion (so-called "super-shedders" and "super-contactors") have been studied extensively; here, we examine the impact of the $g_i - c_i$ interaction as a third mechanism of superspreading, holding $b_i \equiv R_0$ constant and $E_t[c_i(t)] = 1$.

Simulations demonstrate that narrower $g_i(\tau)$ yield more overdispersion in the presence of time-varying contacts. For the periodic contact function with a period of $c_{per} = 1$ day and amplitude $c_{amp} = 0.5$, empirical estimates of k ranged from XX at $\psi = 1$ to XX at $\psi = 0$ for influenza; from XX at $\psi = 1$ to XX at $\psi = 0$ for SARS-CoV-2 omicron; and from XX at $\psi = 1$ to XX at $\psi = 0$ for measles. For the stochastic contact process, when $k_c = 10$ (corresponding to xx), k ranged from XX at $\psi = 1$ to XX at $\psi = 0$ for influenza; from XX at $\psi = 1$ to XX at $\psi = 0$ for SARS-CoV-2 omicron; and from XX at $\psi = 1$ to XX at $\psi = 0$ for measles. This led to meaningful differences in early-outbreak extinction probabilities: for influenza, the extinction probability was XX for $\psi = 0$ *vs.* XX for $\psi = 1$; for SARS-CoV-2 omicron, the extinction probability was XX for $\psi = 0$ *vs.* XX for $\psi = 1$; and for measles, the extinction probability was XX for $\psi = 0$ *vs.* XX for $\psi = 1$.

For the periodic contact model, using results from signal processing theory, we can derive a simple closed-form expression for $\mathrm{OD} = 1/k$ in terms of the model parameters (**Supplementary Methods**):

$$\mathrm{OD} = \frac{c_{amp}^2}{2}\left(1 + \left(\frac{2\pi\bar{g}}{\alpha c_{per}}\right)^2\right)^{-\psi\alpha} \tag{30}$$

where c_{amp} and c_{per} are the amplitude and period of the contact process, respectively. This reveals a few key features about the relationship between ψ and k. First, as expected, $c_{amp} \to 0$ yields $\mathrm{OD} \to 0$, *i.e.*, in the degenerate case where contacts do not vary over time, we recover the standard Poisson-distributed secondary infection distribution with no overdispersion. Also, OD is strictly decreasing in ψ, *i.e.*, narrower infectiousness profiles ($\psi \to 0$) yield more overdispersion. The sensitivity of OD to ψ, however, depends on the magnitude of $\eta = 2\pi\bar{g}/(\alpha c_{per})$. When $\eta \gg 1$, which happens for example when the contact process' period is much shorter than the mean generation time ($c_{per} \ll \bar{g}$), OD is small regardless of ψ. In this case, the contact process varies so quickly that virtually all infectiousness profiles average over the entire contact process, yielding little overdispersion. Alternatively, when $\eta \ll 1$, which can happen when $c_{per} \gg \bar{g}$, $\mathrm{OD} \approx c_{amp}^2/2$ regardless of ψ. In this case, the contact process varies so slowly that all infectiousness profiles, whether narrow or broad, "see" only a small part of the contact process' variation, causing maximal overdispersion that is insensitive to the shape of the individual infectiousness profile.

The impact of ψ is therefore most visible when $\eta \approx 1$, in which case $\psi \to 1$ causes the infectiousness profile to effectively average over the contact variation, while $\psi \to 0$ translates the contact variation into meaningful overdispersion. For our benchmark pathogens, $\bar{g}$ ranges from 3.2 days to 12.2 days. Thus, weekly variation in contacts ($c_{per} = 7$) causes OD to vary meaningfully with ψ. Daily variation ($c_{per} = 1$) gets averaged out, yielding little overdispersion unless $\psi \approx 0$. Monthly variation ($c_{per} = 30$) leads to $\mathrm{OD} \approx c_{amp}^2/2$ regardless of ψ. While the analytic form is less revealing, similar patterns hold for the stochastic contact model, with the switching rate λ and dispersion parameter σ playing analogous roles to c_{per} and c_{amp}.

7 The impact of punctuated infectiousness profiles on controlled epidemics.

7.1 *Detect-and-isolate interventions.*

Detect-and-isolate interventions are among the most important non-pharmaceutical interventions available for curbing an epidemic. The goal of these interventions is to detect individuals at an early stage of infection, prior to peak infectiousness, so that they can shift their behavior (isolate) and prevent onward spread. Detection can happen *via* the development of symptoms and/or diagnostic testing; thus, the timing of symptom onset and the window of test-based detectability, relative to the infectiousness profile, are critical factors influencing the success of such interventions [Fraser].

The efficacy of detect-and-isolate interventions can be summarized in terms of *testing effectiveness* (TE), defined as the expected proportion by which a detect-and-isolate intervention reduces population-level transmission [Middleton and Larremore, 2024]. Formally, TE can be expressed as

$$\mathrm{TE} = \rho\eta P(\tau > D) \tag{31}$$

where $\rho \in [0,1]$ is the participation rate, $\eta \in [0,1]$ is the effectiveness of isolation (so that post-detection transmission attempts succeed with probability $1-\eta$), τ is the time of a transmission attempt relative to the index case's infection time, and D is the detection time. When D is anchored to the index case's infection time (*e.g.*, the detectability window spans 2–7 days after infection), TE depends only on the shape of the generation interval distribution; the shape of the individual infectiousness profile integrates out (**Supplementary Methods**). If, however, D is anchored to the individual infectiousness profile (*e.g.*, the window of detectability extends $\pm$ 3 days from peak infectiousness), TE depends on the width of the individual infectiousness profile, as we illustrate here. Anchoring D to the infectiousness profile is arguably the more realistic scenario since symptom onset, test-based detectability, and peak pathogen shedding are all linked to viral load, which in turn shapes the individual's infectiousness trajectory [Boyer].

To illustrate how TE depends on the shape of the individual infectiousness profile, we considered two detection mechanisms: symptom-based isolation and regular screening with a diagnostic test. For the symptom-based detection mechanism, we assumed symptoms arose at a Normally-distributed time relative to peak infectiousness with mean offset δ_{symp} and standard deviation σ_{symp}, with isolation occurring immediately upon developing symptoms. For the testing-based strategy, we assumed the window of detectability ran from δ_{pre} days before peak infectiousness to δ_{post} days after peak infectiousness, and that diagnostic tests were administered at even intervals of length Δ_{test} with uniform-random offset relative to the start of the detectability window. We included a $\mathrm{Exp}(\lambda_{act})$-distributed test turnaround delay before isolation. To cover various widths of the infectiousness profile, we considered $\psi \in \{0, 0.2, 0.5, 0.8, 1\}$. For each scenario, we numerically integrated the expression $\mathrm{TE} = \eta\rho \int_{-\infty}^{\infty} (1 - F_\epsilon(t)) f_{\mathrm{det}}(t)\, dt$, where F_ϵ is the CDF of the infectiousness kernel ($f_\epsilon \sim \mathrm{Gamma}(\psi\alpha, \beta)$) and f_{det} is the PDF of the detection time distribution, with time indexed from the peak of the individual infectiousness profile.

These experiments demonstrate that narrow infectiousness profiles ($\psi \to 0$) make TE sensitive to small shifts in detection time. When symptoms arise at a $\mathrm{Normal}(\delta_{\mathrm{symp}}, \sigma^2_{\mathrm{symp}})$-distributed time after peak infectiousness, the drop in TE as δ_{symp} shifts from negative to positive (average detection happens before *vs.* after peak infectiousness) is sharpest for narrow infectiousness profiles (**Figure XX**). When $\sigma_{\mathrm{symp}} = 0.5$ and $\delta_{\mathrm{symp}} = 3$, TE ranged from XX for $\psi = 1$ to XX for $\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, XX for $\psi = 1$ to XX for $\psi = 0$ against measles. When $\delta_{\mathrm{symp}} = -1$, the trends reversed: TE ranged from XX for $\psi = 1$ to XX for $\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, XX for $\psi = 1$ to XX for $\psi = 0$ against measles. Similarly, when the window of test-based detectability extended from $\delta_{\mathrm{pre}} = 3$ days prior to peak infectiousness to $\delta_{post} = 7$ days after, and when testing occurred weekly with an Exp(0.5)-distributed turnaround time, TE ranged from XX for $\psi = 1$ to XX for $\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, and XX for $\psi = 1$ to XX for $\psi = 0$ against measles. When testing occurred daily, TE ranged from XX for $\psi = 1$ to XX for

$\psi = 0$ against influenza, XX for $\psi = 1$ to XX for $\psi = 0$ against SARS-CoV-2 omicron, and XX for $\psi = 1$ to XX for $\psi = 0$ against measles. Incomplete participation and imperfect isolation scale these effects, but the qualitative patterns remain (**Supplementary Figure XX**). In short, these findings reveal how narrower profiles cause detect-and-isolate protocols to become increasingly "all-or-nothing", where a detection prior to peak infectiousness averts a person's entire infectious potential while a detection after peak infectiousness averts none.

Beyond the direct reduction in transmission captured by TE, detect-and-isolate interventions produce two additional ψ-dependent effects that act in opposite directions: generation interval truncation (which can accelerate epidemic growth) and increased overdispersion in secondary infection counts (which increases the probability of epidemic extinction). First, detect-and-isolate interventions re-shape the generation interval distribution and cause the mean generation time to contract — an effect that is most pronounced for smooth profiles and disappears as $\psi \to 0$. Under a detect-and-isolate protocol, the generation interval distribution becomes (**Supplementary Methods**)

$$g^*(\tau) = \frac{(1-\eta\sigma)g(\tau) + \eta\sigma \int_0^{\min(\tau,D)} f_I(\tau-\epsilon) f_\epsilon(\epsilon)\, d\epsilon}{1 - \mathrm{TE}} \tag{32}$$

For smooth infectiousness profiles, detection reliably cuts off the tail of transmission; any surviving transmission attempts thus fall earlier in expectation, shortening the mean generation interval (**Figure XX**). For punctuated profiles, the all-or-nothing nature of detection causes some individuals' infectiousness to be cut off entirely, while others are fully missed — and, among those that are missed, their secondary infection times follow $g^*(\tau) = g(\tau)$ on average, yielding no shortening of the mean generation interval. Since shorter mean generation times with fixed R_{eff} yield faster epidemic growth [Wallinga and Lipsitch, 2007], epidemics controlled by detect-and-isolate interventions are expected to grow faster than those simulated "naively" with reproduction number $R_0(1-TE)$ and generation interval $g(\tau)$, with the smoothest individual infectiousness profiles yielding the greatest discrepancies.

Second, detect-and-isolate interventions also increase overdispersion in secondary infection counts, with punctuated infectiousness profiles yielding the most pronounced increases in overdispersion. In the $\psi \to 0$ limit, the variance of the individual reproduction number ν_i is maximized, with

$$\nu_i = \begin{cases} 0 & \text{w.p.TE} \\ R_0 & \text{w.p.}1 - \mathrm{TE} \end{cases} \tag{33}$$

i.e., a distribution consisting of two point masses at 0 and R_0. Overdispersion increases the probability of early-epidemic extinction, so epidemics controlled by detect-and-isolate interventions are expected to go extinct more frequently than those simulated naively with matching reproduction number $R_0(1-TE)$ and generation interval $g(\tau)$, with the most punctuated individual infectiousness profiles yielding the highest extinction probability.

Simulations confirm these expected trends. We simulated epidemics under detect-and-isolate interventions, assuming 50% participation ($\rho = 0.5$), perfect isolation ($\eta = 1$), and a deterministic isolation time 1 day before peak infectiousness (equivalent to $\delta_{\text{symp}} = -1$ and $\sigma_{\text{symp}} = 0$). We also simulated an analogous set of epidemics generated by setting $R_{\text{eff}} = (1-\mathrm{TE})R_0$, but keeping the generation interval $g(\tau)$ unchanged and introducing no overdispersion ($\nu_i \equiv R_0$). We ran 5,000 epidemic simulations in a population of size 10,000 for each of $\psi \in \{0, 0.5, 1\}$ and for each of the three benchmark pathogens. For the detect-and-isolate simulations, the mean ([5% quantile, 95% quantile]) time to reach 500 infections was XX

[XX, XX] for influenza, XX [XX, XX] for SARS-CoV-2 omicron, and XX [XX, XX] for measles; while for the naive TE-adjusted simulations the time to reach 500 infections was XX [XX, XX] for influenza, XX [XX, XX] for SARS-CoV-2 omicron, and XX [XX, XX] for measles. The proportion of epidemics that never reached 500 cases in the detect-and-isolate simulations was XX for influenza, XX for SARS-CoV-2 omicron, and XX for measles; and in the naive TE-adjusted simulations, the proportions were XX for influenza, XX for SARS-CoV-2 omicron, and XX for measles.

Together, these findings reveal that the impact of a detect-and-isolate intervention on an outbreak's trajectory depends sensitively, and through multiple mechanisms, on the shape of the individual infectiousness profile. For smooth individual infectiousness profiles, the reduction in R_{eff} captured by the testing effectiveness can be partially counteracted by generation interval truncation, which can cause the epidemic to grow faster than one might expect. For punctuated individual infectiousness profiles, the reduction in R_{eff} may be compounded by elevated overdispersion in secondary case counts, so that epidemics are not only less intense, but also less likely to establish in the first place.

7.2 *Gathering size restrictions.*

Gathering size restrictions are another important class of non-pharmaceutical interventions, aimed at limiting opportunities for major transmission events. Gathering size restrictions truncate the "tail" of the interpersonal contact distribution, reducing the mean contact rate (and thus the effective reproduction number) as well as reducing overdispersion. Since the individual infectiousness profile's shape is connected with overdispersion in the presence of time-varying contacts, there is reason to believe that gathering size restrictions may play out differently for punctuated *vs.* smooth individual infectiousness profiles. Here, we show that gathering size restrictions yield two main effects: a reduction in mean transmission that is independent of the infectiousness profile's shape, and a reduction in overdispersion whose magnitude depends on ψ.

To investigate this, we considered a modification of the stochastic contact process. As before, the contact level $c_i(t)$ for person i switches at $\text{Exp}(\lambda)$-distributed times, but the new levels are drawn from a truncated $\text{Gamma}(\sigma, \sigma)$ distribution, rather than from the full distribution. The truncation threshold, $c_{\max}$, captures the severity of the restriction.

Truncation reduces the effective reproduction number by the same factor regardless of the individual infectiousness profile's shape. The effective reproduction number is

$$R_{\text{eff}} = \mu_T R_0 \qquad \text{where} \qquad \mu_T = \frac{F_{\text{Gamma}}(c_{\max}; \sigma + 1, \sigma)}{F_{\text{Gamma}}(c_{\max}; \sigma, \sigma)} \tag{34}$$

where $F_{\text{Gamma}}(c; a, b)$ is the $\text{Gamma}(a, b)$ CDF evaluated at c.

It is possible to derive simple expressions for the relative and absolute difference in overdispersion between the uncontrolled and the gathering-size-restricted scenarios (**Supplementary Methods**). These are

$$\frac{\text{OD}_{\text{new}}}{\text{OD}_{\text{old}}} = \frac{\sigma}{\mu_T^2 / V_T} \qquad \text{and} \qquad \text{OD}_{\text{new}} - \text{OD}_{\text{old}} = \text{OD}_{\text{old}} \left[\frac{\sigma}{\mu_T^2 / V_T} - 1 \right] \tag{35}$$

where V_T is the variance of the $\text{Gamma}(\sigma, \sigma)$ distribution truncated at $c_{\max}$. The relative difference in overdispersion does not depend on the individual infectiousness profile's shape; however, the absolute difference does, insofar as OD_{old} depended on the punctuation parameter ψ. Concretely: if OD_{old} is small (≈ 0), which is more likely for smooth individual infectiousness profiles, then $\text{OD}_{\text{new}} - \text{OD}_{\text{old}} \approx 0$, *i.e.*, gathering size restrictions cause little practical difference in an already-small overdispersion. If, however, $\mathrm{OD}_{\mathrm{old}}$ is large, which is more likely for punctuated individual infectiousness profiles, the absolute difference in overdispersion may be meaningful: gathering size restrictions may cause a previously large $\mathrm{OD}_{\mathrm{old}}$ to shrink to a meaningfully smaller $\mathrm{OD}_{\mathrm{new}}$.

To investigate the impact of gathering size restrictions on outbreak trajectories, we simulated 5,000 epidemics in a population of size 10,000 using the stochastic contact model with $\sigma = XX$ and with gathering sizes truncated at $c_{\max} = XX$. We also simulated two parallel sets of "naive" epidemic simulations, both with $R_{\mathrm{eff}} = \mu_T R_0$: one with the full (non-truncated) stochastic contact model, and one with time-uniform contacts ($c_i(t) \equiv 1$). We considered $\psi \in \{0, 0.5, 1\}$ for the three benchmark pathogens. Across simulations, the R_{eff}-adjusted, stochastic contact model over-estimated the epidemic extinction probability and the R_{eff}-adjusted, time-uniform contact model under-estimated it relative to the truncated contact model, and the discrepancies were largest for the narrowest individual infectiousness profiles ($\psi \to 0$) (**Figure XX**). In the $\psi = 0$ boundary case, the proportion of simulated influenza epidemics that failed to establish was XX for the truncated contact model, XX for the R_{eff}-adjusted full stochastic contact model, and XX for the R_{eff}-adjusted time-uniform contact model; for SARS-CoV-2 omicron, XX for the truncated contact model, XX for the R_{eff}-adjusted full stochastic contact model, and XX for the R_{eff}-adjusted time-uniform contact model; and for measles, XX for the truncated contact model, XX for the R_{eff}-adjusted full stochastic contact model, and XX for the R_{eff}-adjusted time-uniform contact model.

Similar to detect-and-isolate interventions, these findings indicate that gathering size restrictions can have additional overdispersion-mediated impacts on epidemic trajectories that go beyond their reduction of mean contact rates, and these impacts are sensitive to the shape of the individual infectiousness profile. Previously, we showed that punctuated profiles generate more overdispersion under time-varying contacts; here, we find that gathering size restrictions attenuate this overdispersion, with the largest absolute reductions occurring precisely when the baseline overdispersion is highest, *i.e.*, with narrow individual infectiousness profiles. Failing to account for this can lead to either over- or under- estimates of the epidemic extinction probability, depending on which model is used.

8 Identifiability and inference of the individual infectiousness profile's shape

Focus here on R_0 and uncertainty in infection timing – that should be enough.

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9 Supplementary Materials

9.1 *Supplementary Methods*

9.1.1 *Deriving population-level quantities from individual-level components*

9.1.2 *Simulation approach*

We used stochastic, individual-based epidemic simulations to investigate the impact of individual-level infectiousness timing on epidemic dynamics. The simulation model has a modular design: an infection attempt generator produces a set of transmission attempts for each infected individual, and a population-level simulation script processes these attempts to propagate the epidemic. We implemented two simulation variants: a finite-population model with susceptible depletion, and an infinite-population model where every infection attempt succeeds. Both are implemented in **R** (version 4.5.0), with performance-critical inner loops reimplemented in C++ via **Rcpp** for speed.

Infection attempt generation. For each newly infected individual i (with infection time t_i), the infection attempt generator proceeds as follows:

1. **Draw the number of infection attempts.** The number of secondary infection attempts χ_i is drawn from a Poisson(b_i) distribution, where b_i is the individual's biological infectiousness. Throughout, we set $b_i \equiv R_0$ for all individuals, so that variation in realized transmission arises solely from infectiousness timing and contact dynamics.

2. **Draw the latent period.** A latent period l_i is drawn from a Gamma$((1-\psi)\alpha, \beta)$ distribution, representing the delay between infection and the onset of transmissibility. When $\psi = 1$, the latent period is zero; when $\psi = 0$, it accounts for the full variance of the generation interval.

3. **Draw the timing of each infection attempt.** For each attempt $j = 1, \ldots, \chi_i$, an independent jitter ϵ_j is drawn from a Gamma$(\psi\alpha, \beta)$ distribution. The time of infection attempt j relative to the index case's infection time is $\tau_{ij} = l_i + \epsilon_j$. When $\psi = 0$, $\epsilon_j = 0$ and all attempts occur simultaneously at time l_i; when $\psi = 1$, the ϵ_j are independent draws from the full Gamma(α, β) generation interval distribution.

4. **Thin for time-varying contacts (if applicable).** When a time-varying contact process $c_i(t)$ is active, each infection attempt at calendar time $t_i + \tau_{ij}$ is accepted with probability $c_i(t_i + \tau_{ij})/c_{\max}$, where $c_{\max}$ is the supremum of the contact process, using Poisson thinning. For the periodic contact model (Eq. 20), all individuals share the same deterministic contact function $c_i(t) = 1 - c_{\text{amp}} \cos(2\pi t / c_{\text{per}})$, and $c_{\max} = 1 + c_{\text{amp}}$. For the stochastic contact model, each individual's piecewise-constant contact trajectory is generated by drawing switching times from a Poisson process with rate λ and contact levels from a Gamma(σ, σ) distribution; the contact trajectory is generated over an interval $[0, 5\bar{g}]$ to ensure enough contact levels are drawn to accommodate all infection attempts. Thinning is performed against the realized maximum contact level over that individual's trajectory.

5. **Thin for detect-and-isolate interventions (if applicable).** When a detect-and-isolate intervention is active, an individual-specific detection time D_i is drawn (relative to peak infectiousness l_i) according to the detection mechanism in effect (symptom-based or screening-based). The individual participates

with probability ρ. If participating, all infection attempts occurring after detection ($\tau_{ij} > t_i + D_i$) are removed with probability η, representing the effectiveness of post-detection isolation.

Finite-population simulation. The finite-population simulation tracks susceptible depletion in a population of size N. It proceeds as follows:

1. **Initialization.** A single index case is infected at time $t = 0$. All other $N - 1$ individuals are susceptible.
2. **Event scheduling.** Each infection attempt is stored as a pending event with its scheduled calendar time $t_i + \tau_{ij}$. Events are maintained in a priority queue, ensuring that the earliest pending event is always processed next.
3. **Event processing.** At each step, the earliest event is removed from the queue. A target individual is selected uniformly at random from the population ($N - 1$ individuals excluding the infector). If the target is susceptible, infection occurs: the target is marked as infected, and its own infection attempts are generated and queued. If the target is already infected or recovered, the attempt fails ("wasted" on a non-susceptible).
4. **Termination.** The simulation terminates when the event queue is empty (epidemic extinction) or when a stopping criterion is met (*e.g.*, a specified number of infections or calendar time is reached).

Infinite-population simulation. The infinite-population simulation is a special case in which every infection attempt succeeds (no susceptible depletion). This is equivalent to a branching process approximation of the early epidemic and is useful for studying early-epidemic dynamics without the confounding effect of herd immunity. The implementation is identical to the finite-population version except that step 3 is simplified: every queued event produces a new infection, with no target selection or susceptibility check.

Default parameters. Unless otherwise stated, simulations used a population size of $N = 10{,}000$, with $n_{\text{sim}} = 5{,}000$ replicate simulations per scenario. Epidemic establishment was defined as reaching $0.05N = 500$ cumulative infections. To cover a range of infectiousness profile shapes, we considered $\psi \in \{0, 0.2, 0.5, 0.8, 1\}$.

Benchmark pathogens. For illustration, we focused on three sets of epidemiological parameters reflecting published estimates of R_0 and $g(\tau)$ for influenza, SARS-CoV-2 (omicron), and measles. The generation interval distribution $g(\tau) \sim \text{Gamma}(\alpha, \beta)$ was parameterized using the mean generation time $\bar{g}$ and variance $\text{Var}[g]$ reported in the literature, with $\alpha = \bar{g}^2/\text{Var}[g]$ and $\beta = \bar{g}/\text{Var}[g]$.

Pathogen	R_0	$\bar{g}$ (days)	$\text{Var}[g]$ (days^2)	α	β
Influenza	2	3.2	4.41	2.32	0.73
SARS-CoV-2 (omicron)	6	7.05	20.8	2.39	0.34
Measles	12	12.2	13.10	11.36	0.93

Table 1: Benchmark pathogen parameters used in simulations.

9.1.3 *Overdispersion in daily infection counts from punctuated infectiousness profiles*

Here we derive the index of dispersion of the daily infection count attributable to a single infector, show that it is monotonically decreasing in the punctuation parameter ψ, and show that the overdispersion aggregates to the population level.

Setup. Fix an index case i, infected at time t_i, who generates $\chi_i \sim \text{Poisson}(R_0)$ total secondary infections. Under the Gamma time-shift model, the generation interval for each secondary infection is $\tau = l_i + \epsilon$, where $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$ is a shared latent period and $\epsilon \sim \text{Gamma}(\psi\alpha, \beta)$ is an independent individual jitter. Let $N_i(\tau)$ be the number of person i's secondary infections that fall on day-since-infection $[\tau, \tau+1)$, and define the random landing probability

$$Q_i(\tau) = \int_\tau^{\tau+1} f_\epsilon(u - l_i)\, du \tag{36}$$

where f_ϵ is the Gamma$(\psi\alpha, \beta)$ density. The randomness in $Q_i(\tau)$ arises from the draw of l_i.

Mean of $Q_i(\tau)$. Taking the expectation over l_i and applying Fubini's theorem:

$$E[Q_i(\tau)] = \int_\tau^{\tau+1} E_{l_i}[f_\epsilon(u - l_i)]\, du = \int_\tau^{\tau+1} (f_l * f_\epsilon)(u)\, du = \int_\tau^{\tau+1} g(u)\, du \equiv q(\tau) \tag{37}$$

since $f_l * f_\epsilon = g$ by construction. Thus $E[Q_i(\tau)]$ depends only on the population-level generation interval distribution and is independent of ψ.

Index of dispersion of $N_i(\tau)$. Conditional on $Q_i(\tau)$, the χ_i offspring are independently assigned to day $[\tau, \tau+1)$ with probability $Q_i(\tau)$. By the Poisson thinning property (marginalizing over χ_i):

$$N_i(\tau) \mid Q_i(\tau) \sim \text{Poisson}(R_0\, Q_i(\tau)) \tag{38}$$

Applying the law of total variance:

$$\begin{aligned} \text{Var}[N_i(\tau)] &= E[\text{Var}[N_i(\tau) \mid Q_i(\tau)]] + \text{Var}[E[N_i(\tau) \mid Q_i(\tau)]] \\ &= R_0\, E[Q_i(\tau)] + R_0^2\, \text{Var}[Q_i(\tau)] \end{aligned} \tag{39}$$

Since $E[N_i(\tau)] = R_0\, E[Q_i(\tau)] = R_0\, q(\tau)$, the index of dispersion is

$$\text{ID}[N_i(\tau)] = \frac{\text{Var}[N_i(\tau)]}{E[N_i(\tau)]} = 1 + R_0\, \frac{\text{Var}[Q_i(\tau)]}{E[Q_i(\tau)]} \tag{40}$$

Extreme cases. In the smooth case ($\psi = 1$), $f_\epsilon = g$ and $l_i = 0$ deterministically, so $Q_i(\tau) = q(\tau)$ is constant across individuals. Thus $\text{Var}[Q_i(\tau)] = 0$ and $\text{ID}[N_i(\tau)] = 1$.

In the spike case ($\psi = 0$), f_ϵ is a point mass at zero. Then $Q_i(\tau) = \mathbf{1}(l_i \in [\tau, \tau+1))$, which is Bernoulli with parameter $q(\tau)$. Thus $\text{Var}[Q_i(\tau)] = q(\tau)(1 - q(\tau))$ and

$$\text{ID}[N_i(\tau)] = 1 + R_0(1 - q(\tau)) \tag{41}$$

Parameter structure

The punctuated Gamma model has four epidemiological parameters: the basic reproduction number R_0, the early-epidemic growth rate r, and the two parameters of the generation interval distribution, α (shape) and β (rate), which together determine the mean generation time $\bar{g} = \alpha/\beta$. These four parameters are not all free: the Euler–Lotka equation

$$1 = R_0 \int_{\tau=0}^{\infty} e^{-r\tau} \frac{\beta^\alpha}{\Gamma(\alpha)} \tau^{\alpha-1} e^{-\beta\tau} d\tau = R_0 \left(\frac{\beta}{\beta + r} \right)^\alpha \tag{42}$$

provides one constraint among them, leaving three free parameters. We choose to work with $(R_0, \bar{g}, \alpha)$ as the free parameters, since each has a direct epidemiological interpretation. The rate parameter β follows immediately from the definition of the mean generation interval:

$$\beta = \frac{\alpha}{\bar{g}}, \tag{43}$$

and the early-epidemic growth rate r follows from the Euler–Lotka equation:

$$r = \beta \left(R_0^{1/\alpha} - 1 \right) = \frac{\alpha}{\bar{g}} \left(R_0^{1/\alpha} - 1 \right). \tag{44}$$

The punctuation parameter $\psi \in [0, 1]$ is the only remaining free parameter, and it is the sole quantity that is not identifiable from population-level epidemic data: since $A(\tau)$ is invariant across ψ by construction, the parameters $(R_0, \bar{g}, \alpha)$ can in principle be estimated from the observed epidemic trajectory independently of ψ. The consequences of varying ψ while holding $(R_0, \bar{g}, \alpha)$ fixed are therefore attributable purely to changes in the individual-level temporal structure of infectiousness, with no confounding from changes in the population-level transmission kernel.

Derivation of $E[\varsigma]$ and $Var[\varsigma]$

Setup. Following Morris et al. [2024], we describe the cumulative infections during the early-epidemic exponential growth phase as a branching process, $Z(t)$. Let r be the epidemic's exponential growth rate; then, as $t \to \infty$, $e^{-rt}Z(t) \to W$ almost surely, where W is a random variable that effectively scales the branching process' initial condition, capturing stochastic fluctuations in the early-epidemic process. We denote this convergence as $Z(t) \approx W \cdot e^{rt}$.

With a change of variables, we can translate W into a random initial time shift:

$$Z(t) \approx e^{r(t+\varsigma)} \qquad \text{where} \qquad \varsigma = \frac{\log W - \log E[W]}{r} = \frac{\log W}{r} \tag{45}$$

when $I_0 = E[W] = 1$ (*i.e.*, when the epidemic is seeded with a single infectious person). Our goal is to determine how $E[\varsigma]$ and $\mathrm{Var}[\varsigma]$ depend on the punctuation parameter ψ. To achieve this, we begin by examining the distribution of W.

Approximating the distribution of W. First, we note that the distribution of W consists of a point mass at 0 with mass q (the probability of epidemic extinction), combined with a smooth density with positive support. The point mass translates into a time shift of $\varsigma = \log(0)/r = -\infty$, corresponding to an epidemic that never establishes. Thus, we restrict our attention to $W|\text{surv}$ and $\varsigma|\text{surv}$, that is, W and ς conditioned on epidemic survival.

To approximate the distribution of $W|\text{surv}$, we use a $\mathrm{Gamma}(k, \lambda)$ distribution matched to the first two moments. This is a version of the five-moment-matching approach described by Morris et al. [2024], simplified for analytic tractability. Specifically, we let

$$k = \frac{E[W|\text{surv}]^2}{\mathrm{Var}[W|\text{surv}]} \qquad \text{and} \qquad \lambda = \frac{E[W|\text{surv}]}{\mathrm{Var}[W|\text{surv}]} \tag{46}$$

Our task is now to compute $E[W|\text{surv}]$ and $\mathrm{Var}[W|\text{surv}]$.

Deriving $E[W|\text{surv}]$. We note that

$$E[W] = E[W|\text{surv}]P(\text{surv}) + E[W|\text{extinct}]P(\text{extinct}) = E[W|\text{surv}] \cdot p \tag{47}$$

since $E[W|\text{extinct}] = 0$, and where $p = 1 - q$ is the probability of epidemic survival. Thus,

$$\boxed{E[W|\text{surv}] = \frac{1}{p}} \tag{48}$$

[for plugging into expressions for k and λ, Eq. 45]

under our assumption that $E[W] = 1$.

Deriving $\mathrm{Var}[W|\text{surv}]$. The derivation for $\mathrm{Var}[W]$ is somewhat more involved. Recall the Gamma time-shift model, where an infectious person produces $\chi_i \sim \mathrm{Poisson}(R_0)$ secondary infections distributed at times $\tau_j = l_i + \epsilon_j$, $j \in \{1, \ldots, \chi_i\}$ and

$$l_i \sim \mathrm{Gamma}((1-\psi)\alpha, \beta), \qquad \epsilon_j \sim \mathrm{Gamma}(\psi\alpha, \beta) \tag{49}$$

This ensures that a random secondary infection time τ is distributed according to the generation interval distribution $g(\tau)$, *i.e.*,

$$\tau \sim \text{Gamma}(\alpha, \beta) \tag{50}$$

For notational convenience, we introduce

$$E[e^{-cr\tau}] = \int_0^\infty e^{-cr\tau} \frac{\beta^\alpha}{\Gamma(\alpha)} \tau^{\alpha-1} e^{-\beta\tau}\, d\tau = \left(\frac{1}{1+cz}\right)^\alpha := \rho_c^\alpha \tag{51}$$

that is,

$$\boxed{\rho_c = \frac{1}{1+cz}} \tag{52}$$

[for notational convenience]

where $z = r/\beta = r\bar{g}/\alpha = R_0^{1/\alpha} - 1$, a dimensionless quantity that simplifies expressions and allows one to easily plug in different combinations of epidemiological parameters. Here, $\bar{g}$ is the mean of the generation interval distribution $g(\tau)$, and c is a positive integer. The final expression, $z = R_0^{1/\alpha} - 1$, comes from the Euler-Lotka equation [Wallinga and Lipsitch, 2007]:

$$1 = R_0 E[e^{-r\tau}] = R_0 \rho_1^\alpha = R_0 \left(\frac{1}{1+z}\right)^\alpha \tag{53}$$

and solving for z.

Next, we introduce the distributional fixed point equation, which is a standard starting point for computing moments of W. This equation exploits the self-similarity of the branching process, in which an epidemic initiated by a single individual is, in distribution, a scaled and time-shifted copy of the whole epidemic. At large t, the epidemic seeded by the index case i consists of the downstream epidemics caused by their offspring j, which have grown approximately to size $Z_j(t) = W_j e^{r(t-\tau_j)}$. That is,

$$Z(t) \approx \sum_{j=1}^{\chi_i} W_j e^{rt} e^{-r\tau_j}$$

Multiplying through by e^{-rt}:

$$W \stackrel{d}{=} \sum_{j=1}^{\chi_i} e^{-r\tau_j} W_j$$

We now use this expression to calculate $\text{Var}[W]$. Since randomness enters through both W and τ, we first condition on the number of offspring and their infection times, $\tau_\bullet = \{\chi_i, \tau_1, \tau_2, \ldots, \tau_{\chi_i}\}$, using the law of total variance:

$$\begin{aligned}
\text{Var}[W] &= E[\text{Var}\Big[\sum_j e^{-r\tau_j} W_j | \tau_\bullet\Big]] + \text{Var}[E\Big[\sum_j e^{-r\tau_j} W_j | \tau_\bullet\Big]] \\
&= E\Big[\sum_j e^{-2r\tau_j} \cdot \text{Var}[W]\Big] + \text{Var}\Big[\sum_j e^{-r\tau_j} \cdot E[W]\Big] \\
&= E\Big[\sum_j e^{-2r\tau_j}\Big] \cdot \text{Var}[W] + (E[W])^2 \cdot \text{Var}\Big[\sum_j e^{-r\tau_j}\Big]
\end{aligned}$$

Since $E[W] = I_0 = 1$, we have

$$\text{Var}[W] = \text{Var}[W]E\Big[\sum_j e^{-2r\tau_j}\Big] + \text{Var}\Big[\sum_j e^{-r\tau_j}\Big]$$

Solving for $\text{Var}[W]$,

$$\text{Var}[W] = \frac{\text{Var}[\sum_j e^{-r\tau_j}]}{1 - E[\sum_j e^{-2r\tau_j}]}$$

To simplify the denominator, note that

$$E\Big[\sum_{j=1}^{\chi_i} e^{-2r\tau_j}\Big] = E[\chi_i]E[e^{-2r\tau_j}] = R_0 E[e^{-2r\tau_j}] = R_0\rho_2^\alpha$$

Thus, the variance expression simplifies to

$$\text{Var}[W] = \frac{\text{Var}[\sum_j e^{-r\tau_j}]}{1 - R_0\rho_2^\alpha}$$

Next, we examine the numerator:

$$\text{Var}[\sum_j e^{-r\tau_j}] = E[(\sum_j e^{-r\tau_j})^2] - E[\sum_j e^{-r\tau_j}]^2 = E[(\sum_j e^{-r\tau_j})^2] - 1$$

where the last step follows from the Euler-Lotka equation (Eq. 52). Then,

$$E[(\sum_j e^{-r\tau_j})^2] = E[\sum_j e^{-2r\tau_j}] + E[\sum_{j\neq k} e^{-r(\tau_j+\tau_k)}]$$

$$= R_0\rho_2^\alpha + E[\chi_i(\chi_i - 1)]E[e^{-r\tau_j}e^{-r\tau_k}]$$

$$= R_0\rho_2^\alpha + R_0^2 E[e^{-r\tau_j}e^{-r\tau_k}]$$

Here, the $\chi_i(\chi_i - 1)$ term enters because there are this many terms in the sum over $j \neq k$. Also,

$$E[\chi_i(\chi_i - 1)] = E[\chi_i^2] - E[\chi_i] = \text{Var}[\chi_i] + E[\chi_i]^2 - E[\chi_i] = R_0 + R_0^2 - R_0 = R_0^2$$

under the assumption that $\chi_i \sim \text{Poisson}(R_0)$.

Since $\tau_j = l_i + \epsilon_j$, and since l_i, ϵ_j, and ϵ_k are independent,

$$E[e^{-r\tau_j}e^{-r\tau_k}] = E[e^{-2rl_i}e^{-r\epsilon_j}e^{-r\epsilon_k}] = E[e^{-2rl_i}](E[e^{-r\epsilon}])^2 = \rho_2^{(1-\psi)\alpha}\cdot\rho_1^{2\psi\alpha}$$

Finally, we can assemble $\text{Var}[W]$:

$$\text{Var}[\sum_j e^{-r\tau_j}] = R_0\rho_2^\alpha + R_0^2\rho_2^{(1-\psi)\alpha}\rho_1^{2\psi\alpha} - 1$$

and thus, substituting in the z expressions for ρ_1 and ρ_2 and noting $R_0 = (1+z)^\alpha$, we obtain

$$\mathrm{Var}[W] = \frac{\left(\frac{1+z}{1+2z}\right)^\alpha + \left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} - 1}{1 - \left(\frac{1+z}{1+2z}\right)^\alpha} := \frac{\sigma_1^\alpha + \sigma_2^{(1-\psi)\alpha} - 1}{1 - \sigma_1^\alpha} \tag{54}$$

where $\sigma_c = \frac{(1+z)^c}{1+2z}$ for positive integer c.

Last, we note that $E[W^n|\mathrm{surv}] = E[W^n]/p$ (by the same logic as in Eq. 46), so

$$\mathrm{Var}[W|\mathrm{surv}] = E[W^2|\mathrm{surv}] - E[W|\mathrm{surv}]^2 = \frac{1}{p}E[W^2] - \frac{1}{p^2}E[W]^2 \tag{55}$$

$$= \boxed{\frac{1}{p}(1 + \mathrm{Var}[W]) - \frac{1}{p^2}} \tag{56}$$

[for plugging into expressions for k and λ, Eq. 45]

Expressions for k and λ. Combining the previous derivations and plugging into Eq. 45 gives

$$\boxed{k = \frac{1 - \sigma_1^\alpha}{p\sigma_2^{(1-\psi)\alpha} - 1 + \sigma_1^\alpha}, \qquad \lambda = pk} \tag{57}$$

[Parameters of the moment-matched Gamma distribution for W]

Deriving the approximate density of ς. Let $f_W(x)$ be the Gamma-distributed approximate density of W:

$$f_W(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x} \tag{58}$$

Since $\varsigma = \frac{1}{r}\log W$, we have $W = e^{r\varsigma}$, or, in terms of the arguments of the density functions, $x = e^{rs}$. By the change of variables formula, the density of ς is

$$f_\varsigma(s) = f_W(e^{rs})|\frac{dw}{ds}| = \frac{\lambda^k}{\Gamma(k)} e^{rs(k-1)} e^{-\lambda e^{rs}} re^{rs} = \frac{r\lambda^k}{\Gamma(k)} e^{rsk} e^{-\lambda e^{rs}} \tag{59}$$

By taking the logarithm, differentiating with respect to s, and setting equal to 0, we can determine the mode of the distribution:

$$\boxed{\mathrm{Mode}[\varsigma] = \frac{1}{r}\log\left(\frac{k}{\lambda}\right) = \frac{1}{r}\log\frac{1}{p}} \tag{60}$$

[Mode of the time shift distribution]

Thus, the modal time shift depends on the epidemic growth rate r and the establishment probability p (and thus the reproduction number R_0), but not on the punctuation parameter ψ.

Deriving $E[\varsigma]$. It would be possible to derive $E[\varsigma]$ directly from Eq.58, but there is a more direct route: since $W \sim \mathrm{Gamma}(k, \lambda)$ and $\varsigma = \frac{1}{r}\log(W)$, it follows that

$$E[\varsigma] = \frac{1}{r}(\Psi^{(0)}(k) - \log\lambda) \tag{61}$$

where $\Psi^{(0)}(k)$ is the digamma function evaluated at k. Plugging in $\lambda = pk$:

$$\boxed{E[\varsigma] = \frac{\log(1/p)}{r} + \frac{\Psi^{(0)}(k) - \log(k)}{r}} \tag{62}$$

[Expected value of the time shift distribution]

The first term is Mode$[\varsigma]$ (Eq. 59), and is independent of the punctuation parameter ψ. The second term is a ψ-dependent adjustment that is:

1. **Strictly negative.** *Proof:* Let $G \sim \text{Gamma}(k, 1)$. Then, $E[G] = k$ and $E[\log G] = \Psi^{(0)}(k)$. By Jensen's inequality, since log is strictly concave,

$$\Psi^{(0)}(k) = E[\log G] < \log(E[G]) = \log k$$

   Thus, $\left(\Psi^{(0)}(k) - \log(k)\right)/r < 0$.

2. **Strictly increasing (less negative) in ψ.** *Proof:* Consider

$$\frac{d}{dk}\left[\Psi^{(0)}(k) - \log k\right] = \frac{d}{dk}\left[\int_0^\infty \left(\frac{e^{-t}}{t} - \frac{e^{-kt}}{1-e^{-t}}\right) dt\right] - \frac{1}{k} = \int_0^\infty \frac{te^{-kt}}{1-e^{-t}}\, dt - \frac{1}{k}$$

   Note that $1 - e^{-t} < t$ for $t > 0$, and thus $\frac{t}{1-e^{-t}} > 1$. So,

$$\int_0^\infty \frac{te^{-kt}}{1-e^{-t}} > \int_0^\infty e^{-kt} = \frac{1}{k}$$

   So,

$$\frac{d}{dk}\left[\Psi^{(0)}(k) - \log k\right] > 0$$

   and thus $(\Psi^{(0)}(k) - \log k)/r$ is increasing in k.
   Next, we note that k is increasing in ψ. The only ψ-dependent quantity in k (Eq.45) is $\sigma_2^{(1-\psi)\alpha}$, which shows up in the denominator. Since

$$\sigma_2 = \frac{(1+z)^2}{(1+2z)} = 1 + \frac{z^2}{1+2z} > 1 \quad \text{for } z > 0$$

   the term is maximized when $\psi = 0$ and minimized when $\psi = 1$. Thus, k is minimized when $\psi = 0$ and maximized when $\psi = 1$ — *i.e.*, k is increasing in ψ.

Together, these observations demonstrate that the expected time shift sits later than the modal time shift (*i.e.*, the time shift is left-skewed; Observation 1), and the expected time shift moves increasingly later as $a_i(\tau)$ becomes more punctuated (Observation 2).

Deriving Var$[\varsigma]$. Following again from the properties of $\log W$ where W is $\text{Gamma}(k, \lambda)$-distributed,

$$\boxed{\text{Var}[\varsigma] = \frac{1}{r^2}\Psi^{(1)}(k)} \tag{63}$$

[Variance of the time shift distribution]

where $\Psi^{(1)}(k)$ is the trigamma function. The trigamma function is strictly decreasing in k, since

$$\frac{d}{dk}\Psi^{(1)}(k) = \frac{d}{dk}\sum_{n=0}^{\infty}\frac{1}{(k+n)^2} = -2\sum_{n=0}^{\infty}\frac{1}{(k+n)^3} \tag{64}$$

which is strictly negative when $k > 0$. Thus, $\psi \to 0$ (punctuated profiles) $\Rightarrow$ k decreasing $\Rightarrow$ $\mathrm{Var}[\varsigma]$ increasing, *i.e.*, more punctuated profiles increase the variance of the time shift.

Summary. In this section, we derived properties of the random time shift ς by relying on a branching process approximation to the early-epidemic exponential growth phase. When conditioning on epidemic survival, we demonstrated that the time shift becomes increasingly later in expectation and more variable as the individual infectiousness profile $a_i(\tau)$ becomes more punctuated ($\psi \to 0$).

Generality beyond the time-shifted Gamma model

The preceding derivation used the time-shifted Gamma model and a Gamma-distributed generation interval. Here, we show that the core result — punctuated profiles increase the variance of early-epidemic timing — holds for *any* non-degenerate generation interval distribution (*i.e.*, any distribution with positive variance, as opposed to a degenerate distribution concentrated on a single point). The argument rests only on the additive splitting $\tau_j = l_i + \varepsilon_j$ (shared shift plus independent jitter) and the Cauchy-Schwarz inequality.

Setup. Let g be an arbitrary generation interval distribution with Laplace transform $M_g(s) = E[e^{-s\tau}]$, so the Euler-Lotka equation gives $R_0 M_g(r) = 1$. Each index case's offspring share a common latent shift l_i and receive independent jitter ε_j, so $\tau_j = l_i + \varepsilon_j$ with the marginal distribution of τ_j equal to g. The distributions of l and ε are arbitrary, subject only to $M_l \cdot M_\varepsilon = M_g$.

Model-free $\mathrm{Var}[W]$ formula. The law-of-total-variance derivation proceeds identically to the Gamma case. The only splitting-dependent step is the cross-term for $j \neq k$:

$$E[e^{-r\tau_j} e^{-r\tau_k}] = E[e^{-2rl}](E[e^{-r\varepsilon}])^2 = M_l(2r) \cdot M_\varepsilon(r)^2$$

Using $M_l(2r) = M_g(2r)/M_\varepsilon(2r)$, this becomes $M_g(2r) \cdot Q$, where

$$Q := \frac{M_\varepsilon(r)^2}{M_\varepsilon(2r)} \tag{65}$$

Assembling the full variance formula (with $A := R_0 M_g(2r)$):

$$\mathrm{Var}[W] = \frac{A(1 + R_0 Q) - 1}{1 - A} \tag{66}$$

Here, A depends only on g and not on the splitting, and Q is the *only* place the splitting enters. Since Eq. 65 is affine in Q with positive slope $R_0 A/(1 - A)$, $\mathrm{Var}[W]$ is strictly increasing in Q.

Three properties of Q.

1. $Q \leq 1$, with equality if and only if ε is degenerate (maximum punctuation).

   Proof: By the Cauchy-Schwarz inequality applied to $X = e^{-r\varepsilon}$:

   $$M_\varepsilon(2r) = E[X^2] \geq (E[X])^2 = M_\varepsilon(r)^2$$

   so $Q = M_\varepsilon(r)^2/M_\varepsilon(2r) \leq 1$. Equality holds if and only if X is almost surely constant, *i.e.*, ε is degenerate. ■

2. Q is multiplicative under convolution: if $\varepsilon_2 \overset{d}{=} \varepsilon_1 + \delta$ with δ independent, then $Q_2 = Q_1 \cdot Q_\delta \leq Q_1$, with strict inequality if δ is non-degenerate.

   Proof: $M_{\varepsilon_2}(s) = M_{\varepsilon_1}(s) \cdot M_\delta(s)$, so $Q_2 = Q_1 \cdot M_\delta(r)^2/M_\delta(2r) = Q_1 \cdot Q_\delta$, and $Q_\delta \leq 1$ by Property 1. ■

3. The spike-vs-smooth gap in $\mathrm{Var}[W]$ is strictly positive for any non-degenerate g.

   Proof: For the spike ($\varepsilon = 0$), $Q_{\text{spike}} = 1$. For the smooth ($\varepsilon = \tau$, $l = 0$), $Q_{\text{smooth}} = M_g(r)^2/M_g(2r) =$

$1/(R_0^2 M_g(2r)) = 1/(R_0 A)$. The gap is

$$\mathrm{Var}[W]_{\mathrm{spike}} - \mathrm{Var}[W]_{\mathrm{smooth}} = \frac{R_0 A(1 - Q_{\mathrm{smooth}})}{1 - A} = \frac{R_0 A - 1}{1 - A}$$

Since $R_0 A = R_0^2 M_g(2r) \geq R_0^2 M_g(r)^2 = 1$ by the Cauchy-Schwarz inequality applied to $X = e^{-r\tau}$, with strict inequality for non-degenerate g, the gap is strictly positive. ■

Monotonicity for parameterized families. For a one-parameter family of splittings ε_ψ with $\varepsilon_0 = 0$ (spike) and $\varepsilon_1 = \tau$ (smooth), if the family is ordered by convolution — meaning $\varepsilon_{\psi_2} \overset{d}{=} \varepsilon_{\psi_1} + \delta$ for some independent $\delta \geq 0$ whenever $\psi_2 > \psi_1$ — then $Q(\psi)$ is strictly decreasing and $\mathrm{Var}[W](\psi)$ is strictly decreasing in ψ. The time-shifted Gamma model satisfies this ordering, since $\mathrm{Gamma}(\psi_2\alpha, \beta) \overset{d}{=} \mathrm{Gamma}(\psi_1\alpha, \beta) + \mathrm{Gamma}((\psi_2 - \psi_1)\alpha, \beta)$, but any infinitely divisible generation interval distribution admits such a family.

Implications for the full chain. Combined with the model-free conditioning step ($\mathrm{Var}[W|\mathrm{surv}] = (1 + \mathrm{Var}[W])/p - 1/p^2$, which increases in $\mathrm{Var}[W]$) and the Gamma-approximation results above, the full chain holds for general g:

$$\begin{aligned}
&\text{More punctuation} \;\Rightarrow\; \underset{\text{(Cauchy-Schwarz)}}{\text{higher } Q} \;\Rightarrow\; \underset{\text{(Eq. 65)}}{\text{higher Var}[W]} \;\Rightarrow\; \underset{\text{(Eq. 45)}}{\text{lower } k} \\
&\Rightarrow\; \underset{\text{(digamma monotonicity)}}{\text{more negative } E[\varsigma]} \;\text{ and }\; \underset{\text{(trigamma monotonicity)}}{\text{higher Var}[\varsigma]}
\end{aligned}$$

The first two links are model-free: the only assumption is that g is non-degenerate. The final link — from $\mathrm{Var}[W]$ to properties of $\varsigma = (\log W)/r$ — requires a distributional assumption, because moments of W do not uniquely determine moments of $\log W$. The Gamma moment-matching approximation serves this role: given $E[W|\mathrm{surv}]$ and $\mathrm{Var}[W|\mathrm{surv}]$, it provides a complete distribution from which $E[\varsigma]$ and $\mathrm{Var}[\varsigma]$ can be computed in closed form via the digamma and trigamma functions. Without this (or an analogous) approximation, the qualitative direction is still supported by Jensen's inequality — which guarantees $E[\varsigma|\mathrm{surv}] < \log(1/p)/r$ for any distribution of $W|\mathrm{surv}$, with the gap increasing when the distribution becomes more dispersed in the sense of convex order — but the specific quantitative expressions would not apply.

Calculating overdispersion

Deriving an expression for the overdispersion parameter k. We begin with the standard assumption that the number of offspring Z_i produced by an index case i is Poisson-distributed:

$$Z_i \sim \mathrm{Poisson}(\nu_i) \tag{67}$$

When ν_i is uniform across people ($\nu_i \equiv R_0$), then the Z_i are truly Poisson-distributed. Overdispersion occurs when ν_i varies across individuals. To capture this overdispersion, the standard approach is to describe Z_i using a Negative Binomial distribution:

$$Z_i \sim \mathrm{NegBin}(R_0, k) \tag{68}$$

which has expectation R_0 and variance $R_0(1 + R_0/k)$. Thus, k captures overdispersion, with $k \to \infty$ yielding

Poisson-distributed Z_i and $k \to 0$ yielding increasingly heavier Negative Binomial tails (*i.e.*, more overdispersion). The Negative Binomial expression is exact when ν_i are Gamma-distributed; otherwise, it is an approximation to the distribution of Z_i. Either way, the parameter k can be obtained *via* moment matching by re-arranging the variance expression:

$$k = \frac{R_0^2}{\mathrm{Var}[Z_i] - R_0} \tag{69}$$

Furthermore, since

$$\mathrm{Var}[Z_i] = \mathrm{Var}[E[Z_i|\nu_i]] + E[\mathrm{Var}[Z_i|\nu_i]] = \mathrm{Var}[\nu_i] + E[\nu_i] = \mathrm{Var}[\nu_i] + R_0 \tag{70}$$

the expression for k simplifies to

$$k = \frac{R_0^2}{\mathrm{Var}[\nu_i]} \tag{71}$$

Next, if we assume everyone's inherent biological infectiousness is the same ($b_i \equiv R_0$), we have

$$\nu_i = R_0 \int_0^\infty g_i(\tau) c_i(t_i + \tau) d\tau := R_0 \tilde{c}_i \tag{72}$$

Plugging in to 70 gives

$$\boxed{k = \frac{1}{\mathrm{Var}[\tilde{c}_i]}} \tag{73}$$

Thus, the overdispersion k depends only on the variance of $\tilde{c}_i = \int_0^\infty g_i(\tau) c_i(t_i + \tau) d\tau$.

Deriving an expression for $\mathrm{Var}[\tilde{c}_i]$. The random variable $\tilde{c}_i$ inherits randomness from both $g_i(\tau)$ and $c_i(t_i + \tau)$. To manage this, we again apply the Law of Total Variance:

$$\begin{aligned}
\mathrm{Var}[\tilde{c}_i] &= \mathrm{Var}[E[\tilde{c}_i|g_i]] + E[\mathrm{Var}[\tilde{c}_i|g_i]] && (74)\\
&= \mathrm{Var}[1] + E[\mathrm{Var}[\tilde{c}_i|g_i]] && (75)\\
&= E[\mathrm{Var}[\tilde{c}_i|g_i]] && (76)
\end{aligned}$$

Thus, we are interested in

$$\tilde{c}_i|g_i = \int_0^\infty g_i(\tau) c_i(t_i + \tau) d\tau \tag{77}$$

where g_i is known (*i.e.*, conditioned upon). This has the form of a classic problem in signal processing: c_i is a wide-sense stationary stochastic process (an input signal); g_i is an impulse response; and $\tilde{c}_i|g_i$ is the output signal evaluated at time t_i. Together, this comprises a linear system. Following standard theory, we can express $\mathrm{Var}[\tilde{c}_i|g_i]$ in terms of the autocorrelation $R(\tau)$ of the signal $\tilde{c}_i|g_i$ at time lag τ. Specifically,

$$\begin{aligned}
\mathrm{Var}[\tilde{c}_i|g_i] &= E[(\tilde{c}_i|g_i)^2] - E[\tilde{c}_i|g_i]^2 && (78)\\
&= R_{\tilde{c}_i|g_i}(0) - E[\tilde{c}_i|g_i]^2 && (79)\\
&= R_{\tilde{c}_i|g_i}(0) - 1 && (80)
\end{aligned}$$

Deriving an expression for $R_{\tilde{c}_i|g_i}(0)$. Deriving an expression for the second moment of the output process, $R_{\tilde{c}_i|g_i}(0)$, is easiest in frequency space. We can introduce $S_\bullet$ as the Fourier transform of $R_\bullet$ and $\hat{g}_i$ as the

Fourier transform of g_i, *i.e.*

$$S_\bullet(\omega) = \int_{-\infty}^{\infty} e^{-i\omega\tau} R_\bullet(\tau) d\tau \qquad \text{and} \qquad \hat{g}_i(\omega) = \int_{-\infty}^{\infty} e^{-i\omega\tau} g_i(\tau) d\tau \tag{81}$$

By the inverse Fourier transform,

$$R_\bullet(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_\bullet(\omega) e^{i\omega\tau} d\omega \tag{82}$$

and thus

$$R_\bullet(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_\bullet(\omega) d\omega \tag{83}$$

We thus seek an expression for $S_{\tilde{c}_i|g_i}(\omega)$, which we can obtain from the fundamental theorem for linear systems:

$$S_{\tilde{c}_i|g_i}(\omega) = |\hat{g}_i(\omega)|^2 S_{c_i}(\omega) \tag{84}$$

Deriving $\hat{g}_i(\omega)$. We begin by examining $\hat{g}_i(\omega)$. Under the Gamma time-shift model, $g_i(\tau) = f_\epsilon(\tau - l_i)$, where f_ϵ is the Gamma$(\psi\alpha, \beta)$ density and $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$. Taking the Fourier transform,

$$\begin{aligned}
\hat{g}_i(\omega) &= \int_{-\infty}^{\infty} g_i(\tau) e^{-i\omega\tau} d\tau && (85)\\
&= \int_{l_i}^{\infty} f_\epsilon(\tau - l_i) e^{-i\omega\tau} d\tau && (86)\\
&= \int_{0}^{\infty} f_\epsilon(\tau) e^{-i\omega(\tau + l_i)} d\tau && (87)\\
&= e^{-i\omega l_i} \int_{0}^{\infty} f_\epsilon(\tau) e^{-i\omega\tau} d\tau && (88)\\
&= e^{-i\omega l_i} \int_{0}^{\infty} \frac{\beta^{\psi\alpha}}{\Gamma(\psi\alpha)} \tau^{\psi\alpha - 1} e^{-\beta\tau} e^{-i\omega\tau} d\tau && (89)\\
&= e^{-i\omega l_i} \Big(\frac{\beta}{\beta + i\omega}\Big)^{\psi\alpha} && (90)
\end{aligned}$$

Taking the squared modulus gives

$$|\hat{g}_i(\omega)|^2 = \Big(1 + \frac{\omega^2}{\beta^2}\Big)^{-\psi\alpha} \tag{91}$$

This is the power transfer function; it is constant across individuals and does not depend on l_i.

Deriving $S_{c_i}(\omega)$ for the periodic contact model. For the periodic contact model

$$c_i(t) = 1 - c_{amp} \cos(\omega_0 t) \tag{92}$$

the autocorrelation is

$$\begin{aligned}
R_{c_i}(\tau) &= E[(1 - c_{amp}\cos(\omega_0 t))(1 - c_{amp}\cos(\omega_0(t+\tau)))] && (93)\\
&= E[1] - c_{amp} E[\cos(\omega_0 t)] - c_{amp} E[\cos(\omega_0(t+\tau))] + c_{amp}^2 E[\cos(\omega_0 t)\cos(\omega_0(t+\tau))] && (94)\\
&= 1 + \frac{c_{amp}^2}{2} \cos(\omega_0 \tau) && (95)
\end{aligned}$$

Taking the Fourier transform,

$$S_{c_i}(\omega) = \int_{-\infty}^{\infty} \Big[1 + \frac{c_{amp}^2}{2}\cos(\omega_0 \tau)\Big] e^{-i\omega\tau} d\tau \tag{96}$$

$$= 2\pi\delta(\omega) + \frac{\pi c_{amp}^2}{2}[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \tag{97}$$

Constructing k for the periodic contact model. Putting everything together, we have:

$$\mathrm{Var}[\tilde{c}_i|g_i] = R_{\tilde{c}_i|g_i}(0) - 1 \tag{98}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{\tilde{c}_i|g_i}(\omega) d\omega - 1 \tag{99}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \Big(1 + \frac{\omega^2}{\beta^2}\Big)^{-\psi\alpha} \Big[2\pi\delta(\omega) + \frac{\pi c_{amp}^2}{2}[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]\Big] d\omega - 1 \tag{100}$$

$$= 1 + \frac{c_{amp}^2}{2}\Big(1 + \frac{\omega_0^2}{\beta^2}\Big)^{-\psi\alpha} - 1 \tag{101}$$

$$= \frac{c_{amp}^2}{2}\Big(1 + \frac{\omega_0^2}{\beta^2}\Big)^{-\psi\alpha} \tag{102}$$

Thus,

$$\boxed{k = \frac{2}{c_{amp}^2}\Big(1 + \frac{\omega_0^2}{\beta^2}\Big)^{\psi\alpha}} \tag{103}$$

Deriving $S_{c_i}(\omega)$ for the Gamma-Poisson contact model. Let $X \sim \mathrm{Gamma}(k_c, k_c)$ be a contact-level draw from the Gamma-Poisson contact model. Then, the autocorrelation

$$R_{c_i}(\tau) = E[c_i(t)c_i(t+\tau)] \tag{104}$$

is equal to $E[X^2] = \mathrm{Var}[X] + E[X]^2 = \frac{1}{k_c} + 1$ with probability $e^{-\lambda|\tau|}$ (*i.e.*, if the contact levels have not switched between times t and $t+\tau$), and is equal to $E[X]^2 = 1$ otherwise. That is:

$$R_{c_i}(\tau) = (\frac{1}{k_c} + 1)e^{-\lambda|\tau|} + (1)(1 - e^{-\lambda|\tau|}) \tag{105}$$

$$= \frac{1}{k_c}e^{-\lambda|\tau|} + 1 \tag{106}$$

Thus:

$$S_{c_i}(\omega) = \int_{-\infty}^{\infty} [\frac{1}{k_c}e^{-\lambda|\tau|} + 1]e^{-i\omega\tau} d\tau \tag{107}$$

$$= \frac{1}{k_c}\int_{-\infty}^{0} e^{\lambda\tau}e^{-i\omega\tau} d\tau + \frac{1}{k_c}\int_{0}^{\infty} e^{-\lambda\tau}e^{-i\omega\tau} d\tau + 2\pi\delta(\omega) \tag{108}$$

$$= \frac{1}{k_c}\Big[\frac{1}{\lambda - i\omega} + \frac{1}{\lambda + i\omega}\Big] + 2\pi\delta(\omega) \tag{109}$$

$$= \frac{2\lambda}{k_c(\lambda^2 + \omega^2)} + 2\pi\delta(\omega) \tag{110}$$

Constructing k for the Gamma-Poisson contact model. Putting everything together, we have:

$$\begin{aligned}
\mathrm{Var}[\tilde{c}_i|g_i] &= R_{\tilde{c}_i|g_i}(0) - 1 && (111)\\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\Big[\frac{2\lambda}{k_c(\lambda^2+\omega^2)} + 2\pi\delta(\omega)\Big]\Big(1+\frac{\omega^2}{\beta^2}\Big)^{-\psi\alpha} d\omega - 1 && (112)\\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty}\Big[\frac{2\lambda}{k_c(\lambda^2+\omega^2)}\Big]\Big(1+\frac{\omega^2}{\beta^2}\Big)^{-\psi\alpha} d\omega && (113)\\
&= \frac{1}{k_c}\int_{-\infty}^{\infty}\frac{\lambda}{\pi(\lambda^2+\omega^2)}\Big(1+\frac{\omega^2}{\beta^2}\Big)^{-\psi\alpha} d\omega && (114)
\end{aligned}$$

and k is the reciprocal of this integral. The integral can be solved in closed form using special functions, but its final form is not especially illuminating; in practice, we evaluate this expression by numerically solving the integral.

9.2 Supplementary Figures